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Grupo: 9
Carrera: Ingeniería en Sistemas de Información
Año: 2023

Análisis de Fourier

Actividad Grupal

Completar los siguientes items:

1) A) Resolver los ejercicios 1 al 4 de la guía práctica.

1. Encuentre la serie de Fourier de ƒ en el intervalo dado:

1. $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 1, & 0 \leq x < \pi \end{cases}$ $\quad T = 2\pi \quad \omega_0 = \frac{2\pi}{T} = 1$

$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\,dt \Rightarrow a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi} 1\,dx\right] = \frac{1}{\pi}\left[x\right]_0^{\pi}$$

$$\boxed{a_0 = 1}$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cos(n\omega_0 t)\,dt \qquad \omega_0 = \frac{2\pi}{2\pi} = 1 \qquad \cos(n\pi) = (-1)^n$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx \Rightarrow a_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi} 1\cos(nx)\,dx\right] = \frac{1}{\pi}\left[\frac{\operatorname{sen} nx}{n}\right]_0^{\pi}$$

$$a_n = \frac{1}{\pi}\left[\frac{\operatorname{sen} n\pi}{n} - \frac{\operatorname{sen} n0}{n}\right] = \frac{1}{\pi}\cdot[0-0] \Rightarrow \boxed{a_n = 0}$$

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\operatorname{sen}(n\omega_0 t)\,dt \Rightarrow b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\operatorname{sen}(nx)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi} 1\operatorname{sen}(nx)\right]$$

$$b_n = \frac{1}{\pi}\left[-\frac{\cos(nx)}{n}\right]_0^{\pi} = \frac{1}{\pi}\left[-\frac{\cos n\pi}{n} + \frac{\cos n0}{n}\right] = \frac{1}{\pi}\cdot\left[-\frac{(-1)^n}{n} + \frac{1}{n}\right]$$

$$\boxed{b_n = \frac{-(-1)^n + 1}{n\pi}}$$

$$f(t) = \frac{1}{2} + \sum_{n=1}^{\infty}\left\{\frac{-(-1)^n + 1}{\pi n}\cdot\operatorname{sen} nx\right\}$$

2. $f(x) = \begin{cases} 1, & -1 < x < 0 \\ x, & 0 < x < 1 \end{cases}$ $\quad T = 2 \quad \omega_0 = \frac{2\pi}{T} = \pi$

$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\,dt \Rightarrow a_0 = 1\int_{-1}^{1} f(x)\,dx = 1\left[\int_{-1}^{0} 1\,dx + \int_{0}^{1} x\,dx\right]$$

$$a_0 = 1\left[x\right]_{-1}^{0} + \left[\frac{x^2}{2}\right]_0^1 = 1\cdot\left(1 + \frac{1}{2}\right) \Rightarrow \boxed{a_0 = \frac{3}{2}}$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cos(n\omega_0 t)\,dt \Rightarrow a_n = 1\int_{-1}^{1} f(x)\cos(n\pi x)\,dx$$

$$a_n = 1\left[\int_{-1}^{0} 1\cdot\cos(n\pi x) + \int_{0}^{1} x\cdot\cos(n\pi x)\right] = \left[\frac{\operatorname{sen}(n\pi x)}{\pi n}\right]_{-1}^{0} + \left[\frac{n\pi x\operatorname{sen}(n\pi x) + \cos(n\pi x)}{n^2\pi^2}\right]_0^1$$

$$a_n = \frac{\operatorname{sen}(n\pi)}{n\pi} + \frac{n\pi\operatorname{sen}(n\pi) + \cos(n\pi) - 1}{n^2\pi^2} \Rightarrow \boxed{a_n = \frac{\operatorname{sen}(n\pi)}{n\pi} + \frac{n\pi\operatorname{sen}(n\pi) + (-1)^n - 1}{n^2\pi^2}}$$

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\operatorname{sen}(n\omega_0 t)\,dt \Rightarrow b_n = 1\int_{-1}^{1} f(x)\operatorname{sen}(n\pi x)\,dx$$

$$b_n = 1\left[\int_{-1}^{0} 1\cdot\operatorname{sen}(n\pi x)\,dx + \int_{0}^{1} x\cdot\operatorname{sen}(n\pi x)\,dx\right] \Rightarrow b_n = \left[-\frac{\cos(n\pi x)}{n\pi}\right]_{-1}^{0} + \left[\frac{\operatorname{sen}(n\pi x) - n\pi x\cos(n\pi x)}{n^2\pi^2}\right]_0^1$$

$$b_n = \frac{\cos(n\pi) - 1}{n\pi} + \frac{\operatorname{sen}(n\pi) - n\pi\cos(n\pi)}{n^2\pi^2} \Rightarrow \boxed{b_n = \frac{(-1)^n - 1}{n\pi} + \frac{\operatorname{sen}(n\pi) - n\pi(-1)^n}{n^2\pi^2}}$$

$$f(t) = 3 + \sum_{n=1}^{\infty}\left[\left(\frac{\operatorname{sen}(n\pi)}{n\pi} + \frac{n\pi\operatorname{sen}(n\pi) + (-1)^n - 1}{n^2\pi^2}\right)\cdot\cos(n\pi x) + \left(\frac{(-1)^n - 1}{n\pi} + \frac{\operatorname{sen}(n\pi) - n\pi(-1)^n}{n^2\pi}\right)\operatorname{sen}(n\pi x)\right]$$

3. $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x^2, & 0 \leq x < \pi \end{cases}$ $\quad T = 2\pi \quad \omega_0 = \frac{2\pi}{T} = 1$

$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\,dt \Rightarrow a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_0^{\pi} x^2\,dx\right] = \frac{1}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi}$$

$$a_0 = \frac{1}{\pi}\cdot\frac{\pi^3}{3} \Rightarrow \boxed{a_0 = \frac{\pi^2}{3}}$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cdot\cos(n\omega_0 t)\,dt \Rightarrow a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_0^{\pi} x^2\cos(nx)\,dx\right]$$

$$a_n = \frac{1}{\pi}\cdot\left[\frac{(n^2x^2-2)\operatorname{sen}(nx) + 2nx\cos(nx)}{n^3}\right]_0^{\pi} \Rightarrow a_n = \frac{(\pi^2n^2-2)\operatorname{sen}(\pi n) + 2\pi n\cos(\pi n)}{\pi n^3}$$

$$\boxed{a_n = \frac{2n(-1)^n}{n^3}}$$

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cdot\operatorname{sen}(n\omega_0 t)\,dt \Rightarrow b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\operatorname{sen}(nx)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_0^{\pi} x^2\operatorname{sen}(nx)\,dx\right]$$

$$b_n = \frac{1}{\pi}\left[\frac{2nx\operatorname{sen}(nx) + (2-n^2x^2)\cos(nx)}{n^3}\right]_0^{\pi} = \frac{2\pi n\operatorname{sen}(\pi n) + (2-\pi^2n^2)\cos(\pi n) - 2}{n^3}$$

$$\boxed{b_n = \frac{(-1)^n(-n^2\pi^2)}{n^3\pi}}$$

$$f(t) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty}\left[\left(\frac{2n(-1)^n}{n^3}\right)\cos(nx) + \left(\frac{(-1)^n(-n^2\pi^2)}{n^3\pi}\right)\operatorname{sen}(nx)\right]$$

4. $f(x) = x + \pi, \quad -\pi < x < \pi \quad T = 2\pi \quad \omega_0 = \frac{2\pi}{T} = 1$

$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\,dt \Rightarrow a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx \Rightarrow a_0 = \int_{-\pi}^{\pi} x + \pi\,dx\cdot\frac{1}{\pi} = \frac{1}{\pi}\left[\frac{x^2}{2} + \pi x\right]_{-\pi}^{\pi}$$

$$a_0 = \frac{1}{\pi}\left[\left(\frac{\pi^2}{2} + \pi^2\right) - \left(\frac{\pi^2}{2} - \pi^2\right)\right] = \frac{1}{\pi}\cdot 2\pi^2 \Rightarrow \boxed{a_0 = 2\pi}$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cos(n\omega_0 t)\,dt \Rightarrow a_n = \frac{1}{\pi}\int_{-\pi}^{\pi}(x+\pi)\cos(nx)\,dx$$

$$a_n = \frac{1}{\pi}\left[\frac{(nx+\pi n)\operatorname{sen}(nx) + \cos(nx)}{n^2}\right]_{-\pi}^{\pi}$$

$$a_n = \frac{1}{\pi}\left[\frac{(n\pi+\pi n)\operatorname{sen}(n\pi) + \cos(n\pi)}{n^2} - \frac{(n(-\pi)+\pi n)\operatorname{sen}(n(-\pi)) + \cos(n(-\pi))}{n^2}\right]$$

$$a_n = \frac{1}{\pi}\,\frac{2\pi\operatorname{sen}(\pi n)}{n} \Rightarrow a_n = \frac{2\pi\operatorname{sen}(\pi n)}{\pi n} \Rightarrow a_n = \frac{2\operatorname{sen}(\pi n)}{n} \Rightarrow \boxed{a_n = 0}$$

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\operatorname{sen}(n\omega_0 t)\,dt \Rightarrow b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}(x+\pi)\operatorname{sen}(nx)\,dx$$

$$b_n = \frac{1}{\pi}\left[\frac{\operatorname{sen}(nx) + (-nx-\pi n)\cos(nx)}{n^2}\right]_{-\pi}^{\pi}$$

$$b_n = \frac{1}{\pi}\left[\frac{\operatorname{sen}(n\pi) + (-n\pi-\pi n)\cos(n\pi)}{n^2} - \frac{\operatorname{sen}(n(-\pi)) + (-n(-\pi)-\pi n)\cos(n(-\pi))}{n^2}\right]$$

$$b_n = \frac{1}{\pi}\cdot\frac{2(\operatorname{sen}(n\pi) - n\pi(-1)^n)}{n^2} \Rightarrow b_n = \frac{2(\operatorname{sen}(n\pi) - n\pi(-1)^n)}{n^2\pi} \Rightarrow \boxed{b_n = \frac{-2n\pi(-1)^n}{n^2\pi}}$$

$$f(t) = \pi + \sum_{n=1}^{\infty}\left[\left(\frac{-2n\pi(-1)^n}{n^2\pi}\right)\operatorname{sen}(nx)\right]$$

5. $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ \operatorname{sen} x, & 0 \leq x < \pi \end{cases}$ $\quad T = 2\pi \quad \omega_0 = \frac{2\pi}{T} = 1$

$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\,dt \Rightarrow a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi}\operatorname{sen} x\,dx\right] = \frac{1}{\pi}\left[-\cos x\right]_0^{\pi}$$

$$a_0 = \frac{1}{\pi}\cdot\left[-\cos(\pi) + \cos(0)\right] = \frac{1}{\pi}\cdot(1+1) \Rightarrow a_0 = \frac{2}{\pi}$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cos(n\omega_0 t)\,dt \Rightarrow a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi}\operatorname{sen}(x)\cos(nx)\right]$$

$$a_n = \frac{1}{\pi}\cdot\left[\frac{n\operatorname{sen}(x)\operatorname{sen}(nx) + \cos(x)\cos(nx)}{n^2-1}\right]_0^{\pi}$$

$$a_n = \frac{1}{\pi}\cdot\left[\frac{n\operatorname{sen}(\pi)\operatorname{sen}(n\pi) + \cos(\pi)\cos(n\pi)}{n^2-1} - \frac{n\operatorname{sen}(0)\operatorname{sen}(n0) + \cos(0)\cos(n0)}{n^2-1}\right]$$

$$a_n = \frac{1}{\pi}\left[\frac{(-1)(-1)^n}{n^2-1} - \frac{1}{n^2-1}\right] = \frac{1}{\pi}\cdot\frac{-(-1)^n+1}{n^2-1} \Rightarrow a_n = -\frac{(-1)^n-1}{\pi(n^2-1)}$$

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\operatorname{sen}(n\omega_0 t)\,dt \Rightarrow b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\operatorname{sen}(nx)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi}\operatorname{sen}(x)\operatorname{sen}(nx)\,dx\right]$$

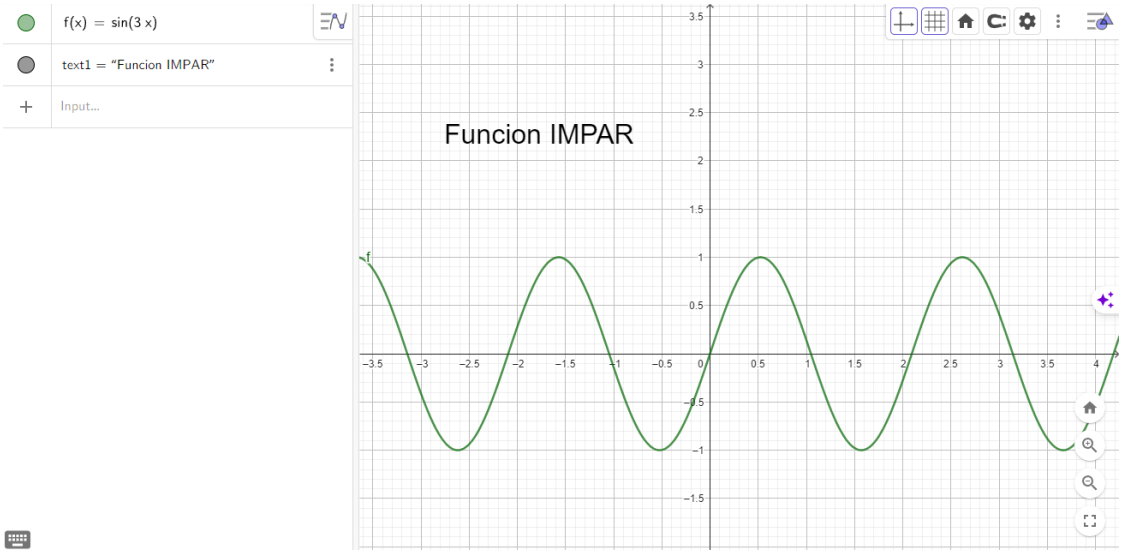
$$b_n = \frac{1}{\pi}\left[\frac{\cos x\operatorname{sen}(nx) - n\operatorname{sen} x\cos(nx)}{n^2-1}\right]_0^{\pi}$$

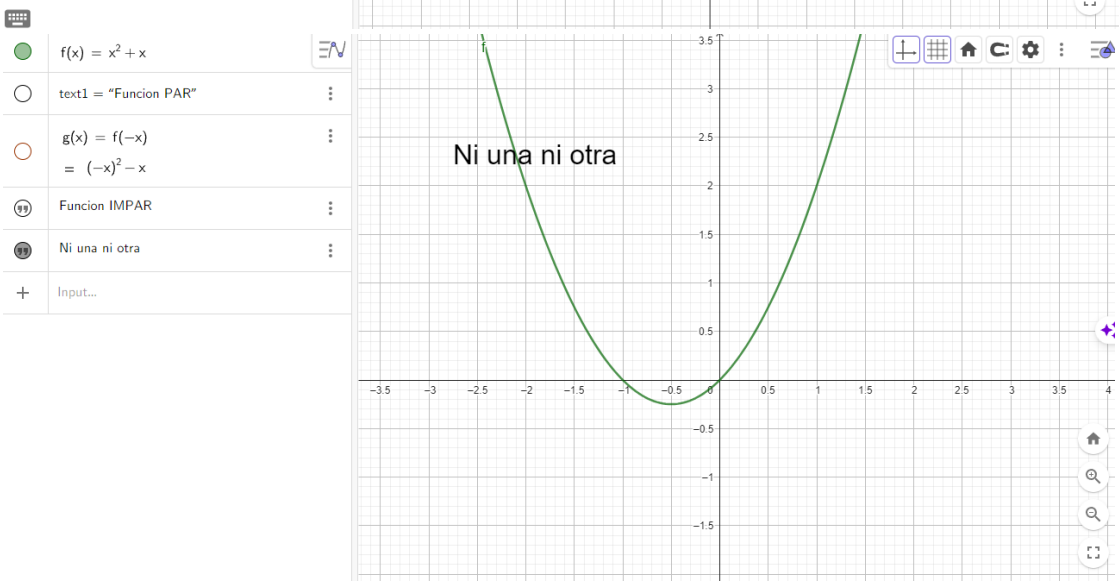
$$b_n = \frac{1}{\pi}\left[\frac{\cos(\pi)\operatorname{sen}(n\pi) - n\operatorname{sen}(\pi)\cos(n\pi)}{n^2-1} - \frac{\cos(0)\operatorname{sen}(n0) - n\operatorname{sen}(0)\cos(n0)}{n^2-1}\right]$$

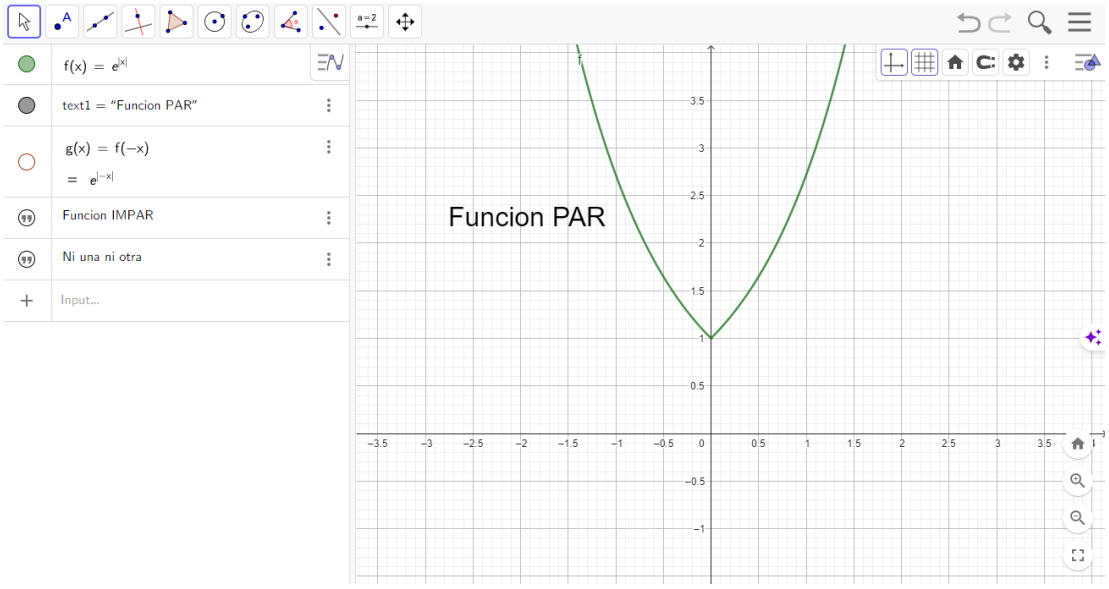
$$b_n = \frac{1}{\pi}\cdot\left[0 - 0\right] \Rightarrow b_n = 0$$

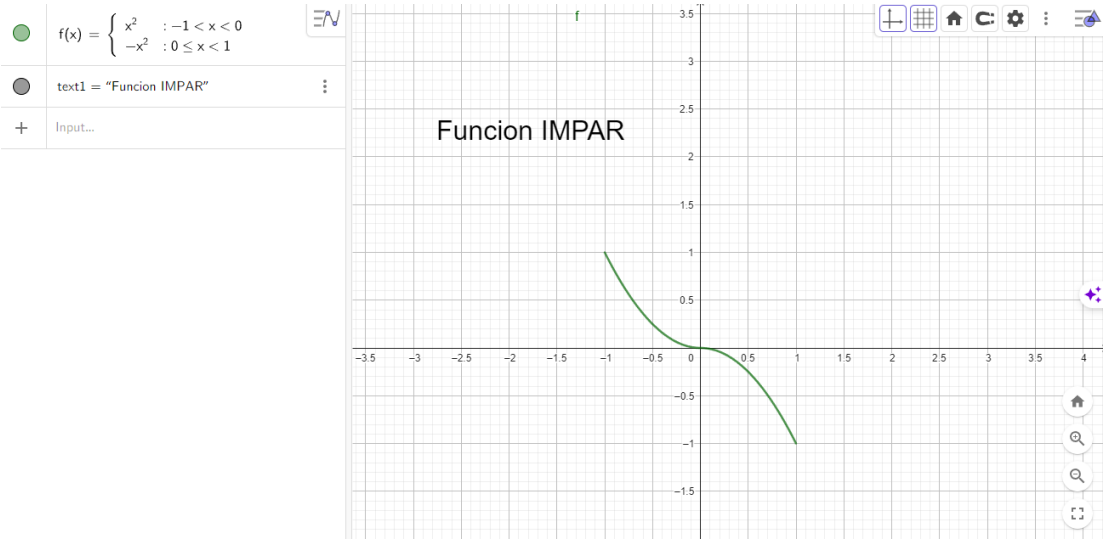
$$f(t) = \frac{1}{\pi} + \sum_{n=1}^{\infty}\left[\left(-\frac{(-1)^n-1}{\pi(n^2-1)}\right)\cdot\cos(nx)\right]$$

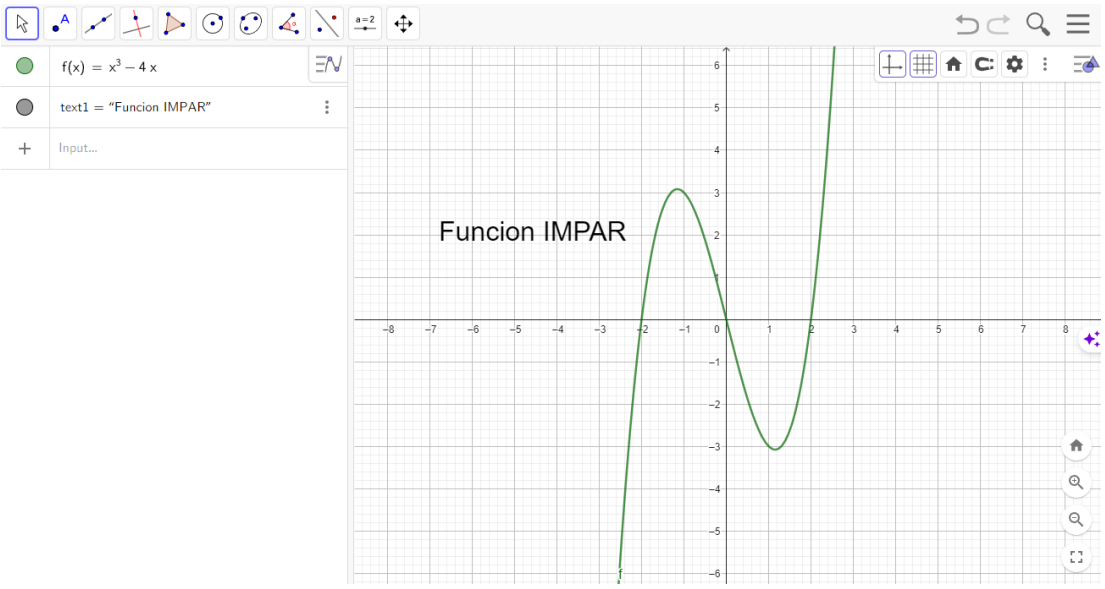
2.

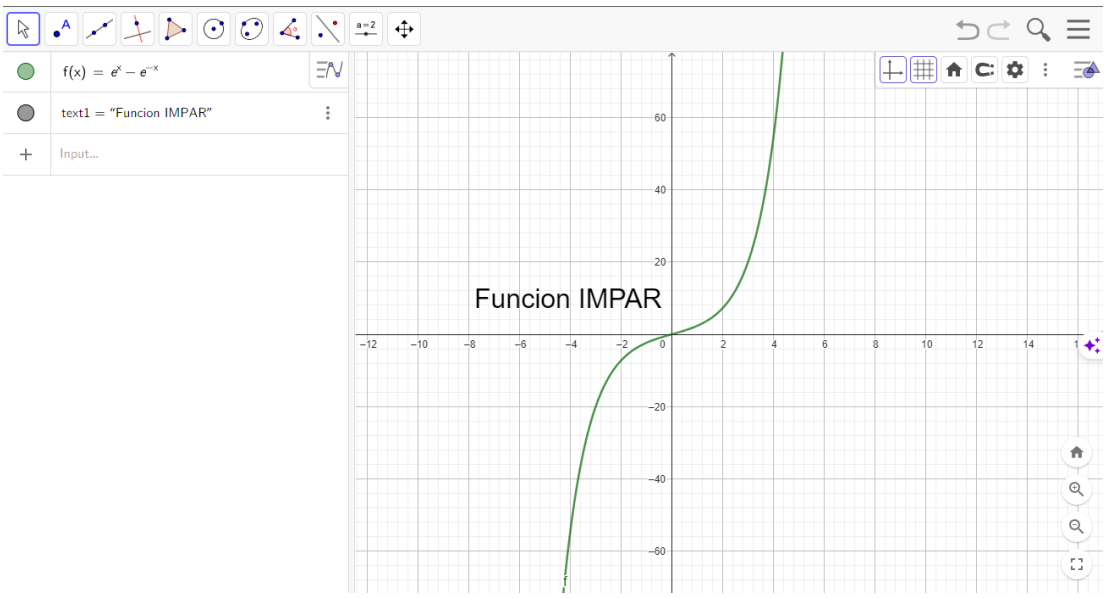

f(x) = sin(3 x)
text1 = "Funcion IMPAR"
Input...
Funcion IMPAR


f(x) = x² + x
text1 = "Funcion PAR"
g(x) = f(−x)
= (−x)² − x
Funcion IMPAR
Ni una ni otra
Input...
Ni una ni otra


f(x) = e^|x|
text1 = "Funcion PAR"
g(x) = f(−x)
= e^|−x|
Funcion IMPAR
Ni una ni otra
Input...
Funcion PAR


f(x) = { x² : −1 < x < 0 ; −x² : 0 ≤ x < 1
text1 = "Funcion IMPAR"
Input...
Funcion IMPAR


f(x) = x³ − 4 x
text1 = "Funcion IMPAR"
Input...
Funcion IMPAR


f(x) = eˣ − e⁻ˣ
text1 = "Funcion IMPAR"
Input...
Funcion IMPAR

3.

1-

$$f(x) = \begin{matrix} -1\ si - \pi < x < 0 \\ 1\ si\ \ 0 \leq x < \pi \end{matrix}$$

$$T = 2\pi$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

$$a_0 = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)dx$$

$$a_0 = \int_{-\pi}^{0} -1dx + \int_{0}^{\pi} 1dx = -x]_{-\pi}^{0} + x]_{0}^{\pi} = -\pi + \pi = 0$$

$$a_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)\cos(n\omega_0 x)\,dx$$

$$a_n = \int_{-\pi}^{0} -1 * \cos(nx)\,dx + \int_{0}^{\pi} 1 * \cos(nx)\,dx$$

$$a_n = \int_{-\pi}^{0} -\cos(nx)\,dx + \int_{0}^{\pi} \cos(nx)\,dx = -\frac{sen(nx)}{n}\Big]_{-\pi}^{0} + \frac{sen(nx)}{n}\Big]_{0}^{\pi} = 0$$

Como la función es impar y el coseno es par, da una función impar resultante cuando se multiplican, y una función impar integrada entre extremos equidistantes del 0 siempre da 0
Se hará una serie de senos para representar la función porque no hay posibilidad de hacerla de cosenos

$$b_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)\,\mathrm{sen}(n\omega_0 x)\,dx$$

$$b_n = \int_{-\pi}^{0} -1 * \mathrm{sen}(nx)\,dx + \int_{0}^{\pi} 1 * \mathrm{sen}(nx)\,dx$$

$$b_n = \int_{-\pi}^{0} -\mathrm{sen}(nx)\,dx + \int_{0}^{\pi} \mathrm{sen}(nx)\,dx = \frac{cos(nx)}{n}\Big]_{-\pi}^{0} + \frac{-cos(nx)}{n}\Big]_{0}^{\pi}$$

$$b_n = \frac{cos(0n)}{n} - \frac{cos(-\pi n)}{n} - \frac{cos(\pi n)}{n} + \frac{cos(0x)}{n} = \frac{1}{n} - \frac{(-1)^n}{n} - \frac{(-1)^n}{n} + \frac{1}{n}$$

$$b_n = \frac{1 - (-1)^n - (-1)^n + 1}{n} = \frac{2 - 2(-1)^n}{n}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2 - 2(-1)^n}{n}\,\mathrm{sen}\,(nx)$$

2-

$$f(x) = |x|\ si - \pi < x < \pi$$

$$f(x) = \begin{matrix} -x\ si & -\pi < x \leq 0 \\ x\ si & 0 < x \leq \pi \end{matrix}$$

$$T = 2\pi$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

$$a_0 = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)dx$$

$$a_0 = \int_{-\pi}^{0} -xdx + \int_{0}^{\pi} xdx = -\frac{x^2}{2}\Big]_{-\pi}^{0} + \frac{x^2}{2}\Big]_{0}^{\pi} = \frac{\pi^2}{2} + \frac{\pi^2}{2} = \pi^2$$

Se hará una serie de cosenos porque al ser una función par $(|x|)$*por una impar* $\big(sen(x)\big)$ *se obtiene una impar, por lo que la integral dará* 0.

$$a_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)\, cos(n\omega_0 x)\, dx$$

$$a_n = \int_{-\pi}^{0} -x * \cos(nx)\, dx + \int_{0}^{\pi} x * \cos(nx)\, dx$$

$$a_n = -\int_{-\pi}^{0} x\cos(nx)\, dx + \int_{0}^{\pi} x\cos(nx)\, dx = 2\int_{0}^{\pi} x\cos(nx)\, dx \;\; (Por\ ser\ función\ par)$$

$$\int_{0}^{\pi} x\cos(nx)\, dx = \frac{xsen(nx)}{n}\Big]_{0}^{\pi} - \int_{0}^{\pi} \frac{sen(nx)}{n} dx = \frac{xsen(nx)}{n}\Big]_{0}^{\pi} + \frac{cos(nx)}{n^2}\Big]_{0}^{\pi}$$

$$= \frac{cos(\pi x)}{n^2} - \frac{cos(0x)}{n^2} = \frac{(-1)^n}{n^2} - \frac{1}{n^2} = \frac{(-1)^n - 1}{n^2}$$

$$a_n = \frac{2(-1)^n - 2}{n^2}$$

$$f(x) = \frac{\pi^2}{2} + \sum_{n=1}^{\infty} \frac{2(-1)^n - 2}{n^2} \cos(nx)$$

3-

$$f(x) = x^2, -1 < x < 1$$

$$T = 2$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

$$a_0 = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)dx$$

$$a_0 = \int_{-1}^{1} x^2 dx = \frac{x^3}{3}\Big]_{-1}^{1} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

Como x^2 *es una función par tenemos que hacer una serie de* cosenos (función par)

$$a_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)\, cos(n\omega_0 x)\, dx$$

$$a_n = \int_{-1}^{1} x^2 \cos(n\pi x)\, dx = \frac{x^2 sen(n\pi x)}{n\pi}\Big]_{-1}^{1} - \int_{-1}^{1} \frac{2x\,\mathrm{sen}(n\pi x)}{n\pi} dx =$$

$$= \frac{x^2 sen(n\pi x)}{n\pi}\Big]_{-1}^{1} - \left[\frac{-2xcos(n\pi x)}{n^2\pi^2}\Big]_{-1}^{1} + \int_{-1}^{1} \frac{2cos(n\pi x)}{n^2\pi^2} dx\right]$$

$$= \frac{x^2 sen(n\pi x)}{n\pi}\Bigg]_{-1}^{1} + \frac{2x cos(n\pi x)}{n^2\pi^2}\Bigg]_{-1}^{1} - \frac{2sen(n\pi x)}{n^3\pi^3}\Bigg]_{-1}^{1}$$

$$= \frac{sen(n\pi)}{n\pi} - \frac{sen(-n\pi)}{n\pi} + \frac{2cos(n\pi)}{n^2\pi^2} + \frac{2cos(-n\pi)}{n^2\pi^2} - \frac{2sen(\pi x)}{n^3\pi^3} - \frac{2sen(-\pi x)}{n^3\pi^3}$$

$$a_n = \frac{2cos(n\pi)}{n^2\pi^2} + \frac{2cos(-n\pi)}{n^2\pi^2} = \frac{2(-1)^n}{n^2\pi^2} + \frac{2(-1)^n}{n^2\pi^2} = \frac{4(-1)^n}{n^2\pi^2}$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2\pi^2} \cos(nx)$$

4-

$$f(x) = \pi^2 - x^2, -\pi < x < \pi$$

$$T = 2\pi$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

$$a_0 = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)dx$$

$$a_0 = \int_{-\pi}^{\pi} (\pi^2 - x^2)dx = \pi^2 x - \frac{x^3}{3}\Bigg]_{-\pi}^{\pi} = \left(\pi^2\pi - \frac{\pi^3}{3}\right) - \left(-\pi^2\pi + \frac{\pi}{3}\right) =$$

$$a_0 = \pi^3 - \frac{\pi^3}{3} + \pi^3 - \frac{\pi^3}{3} = 2\pi^3 - \frac{2\pi^3}{3} = \frac{4\pi^3}{3}$$

Como $\pi^2 - x^2$ es una función par tenemos que hacer una serie de cosenos (función par)

$$a_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)\, cos(n\omega_0 x)\, dx$$

$$a_n = \int_{-\pi}^{\pi} (\pi^2 - x^2)\cos(nx)\, dx = \frac{(\pi^2 - x^2)sen(nx)}{n}\Bigg]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \frac{-2x\, \mathrm{sen}(nx)}{n} dx =$$

$$= \frac{(\pi^2 - x^2)sen(nx)}{n}\Bigg]_{-\pi}^{\pi} - \left[\frac{2x cos(nx)}{n^2}\Bigg]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \frac{2cos(nx)}{n^2} dx\right]$$

$$= \frac{(\pi^2 - x^2)sen(nx)}{n}\Bigg]_{-\pi}^{\pi} - \frac{2x cos(nx)}{n^2}\Bigg]_{-\pi}^{\pi} + \frac{2sen(nx)}{n^3}\Bigg]_{-\pi}^{\pi}$$

$$\frac{(\pi^2 - \pi^2)sen(n\pi)}{n} - \frac{(\pi^2 - (-\pi)^2)sen(-n\pi)}{n} = 0$$

$$\frac{2sen(n\pi)}{n^3} - \frac{2sen(-n\pi)}{n^3} = 0$$

$$a_n = -\left(\frac{2\pi cos(n\pi)}{n^2} + \frac{2\pi cos(-n\pi)}{n^2}\right) = \frac{-4\pi(-1)^n}{n^2}$$

$$f(x) = \frac{4\pi^3}{6} + \sum_{n=1}^{\infty} \frac{-4\pi(-1)^n}{n^2} \cos(nx)$$

5-

$$f(x) = \begin{matrix} x-1 \; si \; -\pi < x < 0 \\ x+1 \quad si \;\; 0 \leq x < \pi \end{matrix}$$

$$T = 2\pi$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

$$a_0 = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)dx$$

$$a_0 = \int_{-\pi}^{0} (x-1)dx + \int_{0}^{\pi} (x+1)dx = \left(\frac{x^2}{2} - x\right)\Bigg]_{-\pi}^{0} + \left(\frac{x^2}{2} + x\right)\Bigg]_{0}^{\pi}$$

$$a_0 = \left(\frac{-\pi^2}{2} + \pi\right) - \left(\frac{\pi^2}{2} + \pi\right) = \frac{\pi^2}{2} + \pi - \frac{\pi^2}{2} - \pi = 0$$

Como es una función impar, vamos a usar una serie de senos porque el producto entre dos funciones impares es una función par, por lo que no se anulará.

$$b_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)\,\mathrm{sen}(n\omega_0 x)\,dx$$

$$b_n = \int_{-\pi}^{0} (x-1) * \mathrm{sen}(nx)\,dx + \int_{0}^{\pi} (x+1) * \mathrm{sen}(nx)\,dx$$

$$= 2\int_{0}^{\pi} (x+1) * \mathrm{sen}(nx)\,dx \;\; (Por\ ser\ función\ par)$$

$$b_n = 2\int_{0}^{\pi} (x+1) * \mathrm{sen}(nx)\,dx = 2\frac{-(\mathrm{x}+1)\cos(nx)}{n}\Bigg]_{0}^{\pi} - 2\int_{0}^{\pi} \frac{\cos(nx)}{n}dx$$

$$b_n = 2\frac{-(\mathrm{x}+1)\cos(nx)}{n}\Bigg]_{0}^{\pi} - 2\frac{sen(nx)}{n^2}\Bigg]_{0}^{\pi}$$

$$b_n = \frac{-2\,(\pi+1)\cos(n\pi)}{n} - \frac{-2\,(1)\cos(0)}{n} = \frac{(-2\pi-2)(-1)^{\mathrm{n}} + 2}{n}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{(-2\pi-2)(-1)^{\mathrm{n}} + 2}{n}\mathrm{sen}\,(nx)$$

4.

1) $f(x) \begin{cases} 1, & 0 < x < 1/2 \\ 0, & 1/2 \leq x < 1 \end{cases}$ $\quad T = 2 \quad L = 1$

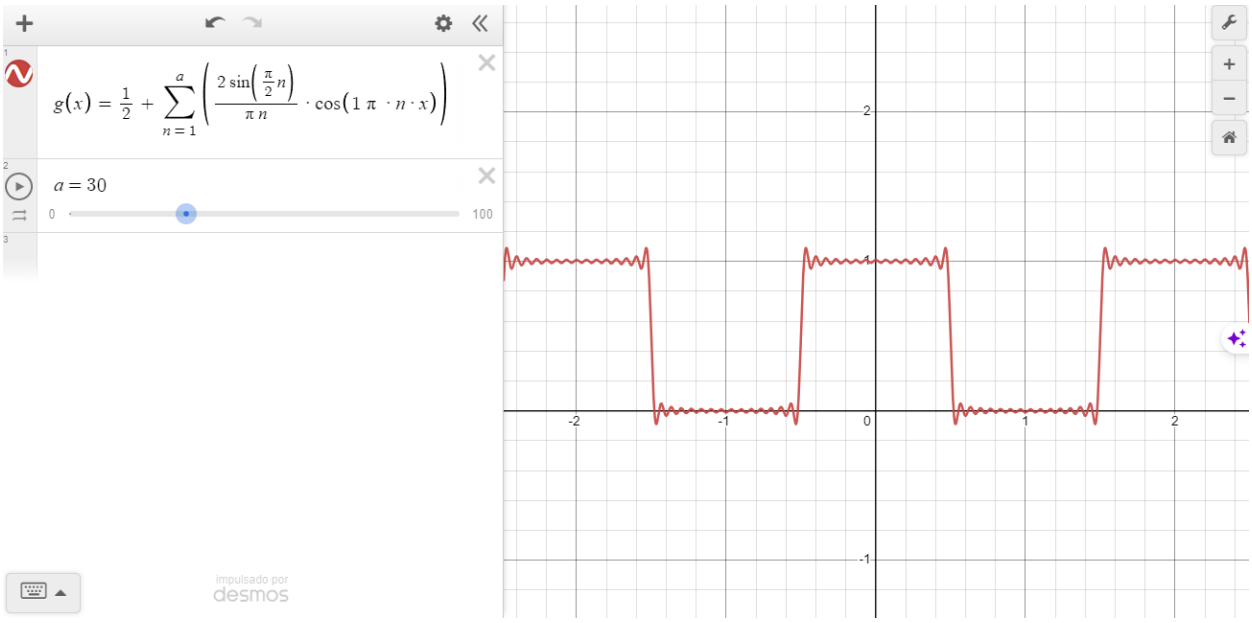
$\omega_0 = \frac{2\pi}{2} = \boxed{\pi}$

$$a_0 = \frac{2}{2}\left(\int_{-1}^{-1/2} 0\,dt + \int_{-1/2}^{1/2} 1\,dt + \int_{1/2}^{1} 0\,dt\right) \qquad a_n = \frac{2}{2}\left(\int_{-1}^{-1/2} 0\,dt + \int_{-1/2}^{1/2} 1 \cdot \cos(\pi n t)\,dt + \int_{1/2}^{1} 0\,dt\right)$$

$$a_0 = 1 \cdot \left(x \Big|_{-1/2}^{1/2} \right) = \boxed{1} \qquad a_n = 1 \cdot \left(\frac{\operatorname{sen}(\pi n x)}{\pi n} \Big|_{-1/2}^{1/2} \right)$$

$$\boxed{b_n = 0} \qquad a_n = \boxed{\frac{2 \operatorname{sen}\left(\frac{\pi n}{2}\right)}{\pi n}}$$

$$\boxed{f(x) = \frac{1}{2} + \sum_{n=1}^{K} \left(\frac{2 \operatorname{sen}\left(\frac{\pi n}{2}\right)}{\pi n} \cdot \cos(n \pi x) \right)}$$



2) $f(x) = \cos x \;,\; 0 < x < \pi/2$

$T = \pi$

$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{\pi} = 2$

$$a_0 = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} \cos(t)\,dt \qquad \boxed{b_n = 0}$$

$$= \frac{2}{\pi}\cdot\left|\operatorname{sen}(x)\right|_{-\pi/2}^{\pi/2}$$

$$= \frac{2}{\pi}\cdot 2 = \boxed{\frac{4}{\pi}}$$

$$\boxed{b_n = 0}$$

$$a_n = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} \cos(t)\cdot\cos(n\cdot 2\cdot t)\,dt$$

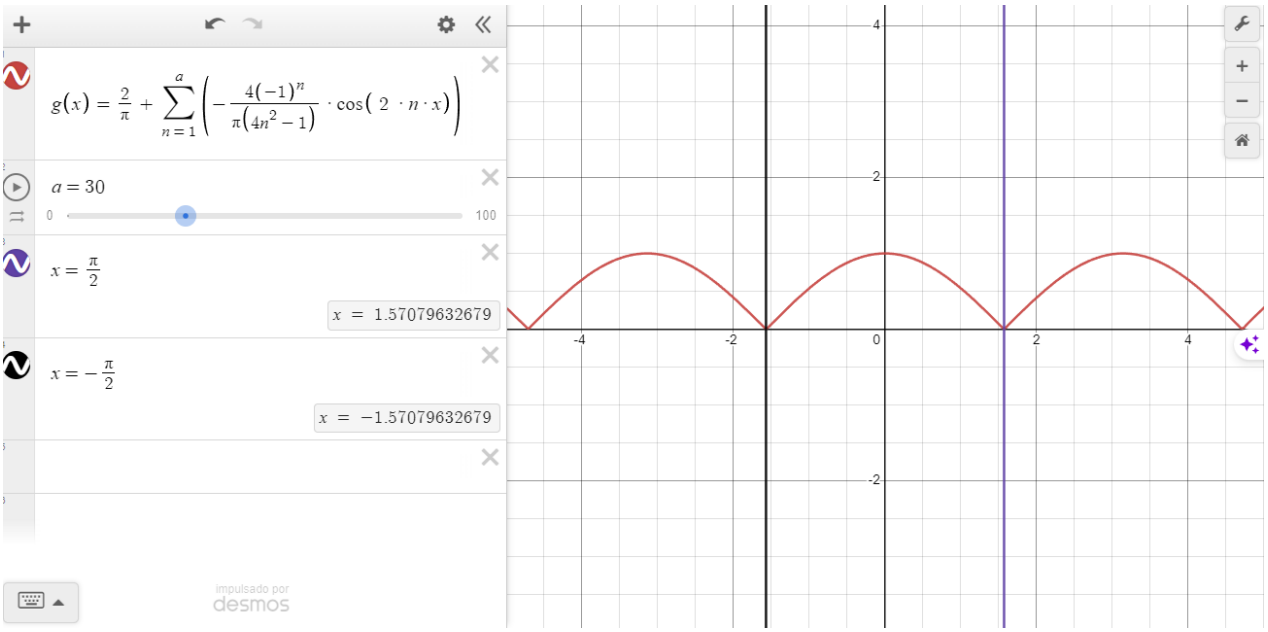
$$a_n = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} \frac{\cos((2n+1)x) + \cos((2n-1)x)}{2}$$

$$a_n = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} \frac{\cos((2n+1)x)}{2} + \int_{-\pi/2}^{\pi/2} \frac{\cos((2n-1)x)}{2}$$

$$a_n = \frac{2}{\pi}\cdot\left(\frac{\operatorname{sen}((2n+1)x)}{2(2n+1)}\Bigg|_{-\pi/2}^{\pi/2} + \frac{\operatorname{sen}((2n-1)x)}{2(2n-1)}\Bigg|_{-\pi/2}^{\pi/2}\right)$$

$$a_n = \frac{2}{\pi}\cdot\left(-\frac{2\cos(\pi n)}{4n^2-1}\right) = \boxed{-\frac{4(-1)^n}{\pi(4n^2-1)}}$$

$$\boxed{f(x) = \frac{2}{\pi} + \sum_{n=1}^{K}\left(\frac{-4(-1)^n}{\pi(4n^2-1)}\cdot\cos(n\,2\,x)\right)}$$



3)

$$f(x) = \begin{cases} x & 0 < x < \pi/2 \\ \pi - x & \pi/2 \le x < \pi \end{cases} \qquad L = \pi \qquad T = 2\pi$$

$$\omega_0 = \frac{2\pi}{2\pi} = \boxed{1}$$

$$a_0 = \frac{2}{L}\left(\int_0^{\pi/2} x\,dx + \int_{\pi/2}^{\pi} \pi - x\,dx\right)$$

$$a_0 = \frac{2}{\pi}\cdot\left(\frac{x^2}{2}\Big|_0^{\pi/2} + \pi x - \frac{x^2}{2}\Big|_{\pi/2}^{\pi}\right)$$

$$a_0 = \frac{2}{\pi}\cdot\left(\frac{\pi^2}{8} + \frac{\pi^2}{8}\right) = \frac{2}{\pi}\frac{\pi^2}{4} = \boxed{\frac{1}{2}\pi}$$

$$a_n = \frac{2}{\pi}\left(\int_0^{\pi/2} t.\cos(nt)\,dt + \int_{\pi/2}^{\pi} (\pi - t).\cos(nt)\,dt\right)$$

$$a_n = \frac{2}{\pi}\left(\frac{2\cos\left(\pi\frac{n}{2}\right) - \cos(\pi n) - 1}{n^2}\right)$$

$$b_n = 0$$

$$\boxed{f(x) = \frac{\pi}{4} + \sum_{n=1}^{K}\left(\frac{2}{\pi}\left(\frac{2\cos\left(\pi\frac{n}{2}\right) - \cos(\pi n) - 1}{n^2}\right).\cos(nt)\right)}$$

aux

$$\int_0^{\pi/2} \underbrace{t}_{u}\,\underbrace{\cos(nt)\,dt}_{dv} \qquad -\int \underbrace{(x-\pi)}_{u}\,\underbrace{\cos(nt)\,dt}_{dv}$$

$$u = x - \pi \qquad dv = \cos(nt)\,dt$$

$$du = 1 \qquad v = \frac{\operatorname{sen}(nt)}{n}$$

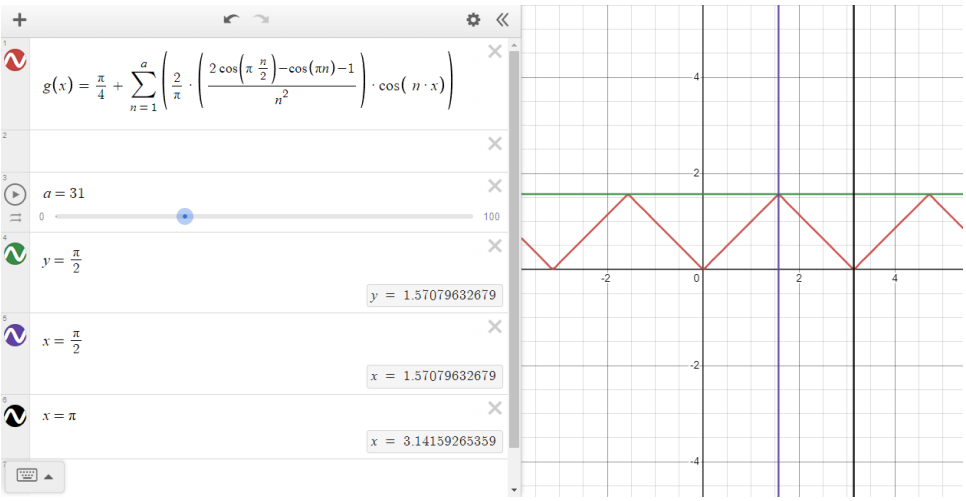
$$u = t \qquad dv = \cos(nt)\,dt$$

$$du = 1 \qquad v = \frac{\operatorname{sen}(nt)}{n}$$

$$= \frac{t\operatorname{sen}(nt)}{n} - \int \frac{\operatorname{sen}(nt)}{n}\,dt \qquad = \frac{(x-\pi)\operatorname{sen}(nt)}{n} - \int \frac{\operatorname{sen}(nt)}{n}\,dt$$

$$= \frac{t\operatorname{sen}(nt)}{n} + \frac{\cos(nt)}{n^2}\Big|_0^{\pi/2} \qquad = \frac{(t-\pi)\operatorname{sen}(nt)}{n} + \frac{\cos(nt)}{n^2}\Big|_{\pi/2}^{\pi}$$

$$= \frac{\pi n\operatorname{sen}\left(\frac{\pi n}{2}\right) + 2\cos\left(\frac{n\pi}{2}\right) - 2}{2n^2} \qquad = \frac{-2\cos(\pi n) + \pi n\operatorname{sen}\left(\frac{\pi n}{2}\right) - 2\cos\left(\frac{\pi n}{2}\right)}{2n^2}$$



4) $f(x) = \begin{cases} x, & 0 < x < 1 \\ 1, & 1 \leq x < 2 \end{cases}$

$L = 2 \qquad T = 4$

$\omega_0 = \frac{2\pi}{4} = \boxed{\frac{1}{2}\pi}$

$a_0 = \frac{2}{2}\left(\int_0^1 t\,dt + \int_1^2 1\,dt\right)$

$a_n = \frac{2}{2}\left(\int_0^1 t\cos\left(n\frac{\pi}{2}t\right)dt + \int_1^2 \cos\left(n\frac{\pi}{2}t\right)dt\right)$

$a_0 = 1\cdot\left(\frac{x^2}{2}\Big|_0^1 + x\Big|_1^2\right)$

$a_n = 1\cdot\left(2\cdot\frac{\pi\,\text{sen}(\pi n) + 2\cos\left(\pi\frac{n}{2}\right) - 2}{n^2\pi^2}\right)$

$a_0 = \frac{1}{2} + 1 = \boxed{\frac{3}{2}}$

$f(x) = \frac{3}{4} + \sum_{n=1}^{K}\left(\frac{2\pi n\,\text{sen}(\pi n) + 2\cos\left(\pi\frac{n}{2}\right) - 2}{n^2\pi^2}\cdot\cos\left(nt\cdot\frac{1}{2}\pi\right)\right)$

$b_n = \boxed{0}$

aux

$\int_0^1 t\cos\left(n\frac{\pi}{2}t\right)dt \Rightarrow 4\int_0^1 u\cos(n\pi u)\,du$

$u = \frac{x}{2} \qquad dx = 2du \qquad f = u \qquad dg' = \cos(n\pi u)$

$du = \frac{1}{2} \qquad df = 1 \qquad g = \frac{\text{sen}(\pi n u)}{\pi n}$

$4\left(\frac{u\,\text{sen}(\pi n u)}{\pi n} - \int\frac{\text{sen}(\pi n u)}{\pi n}\,du\right)$

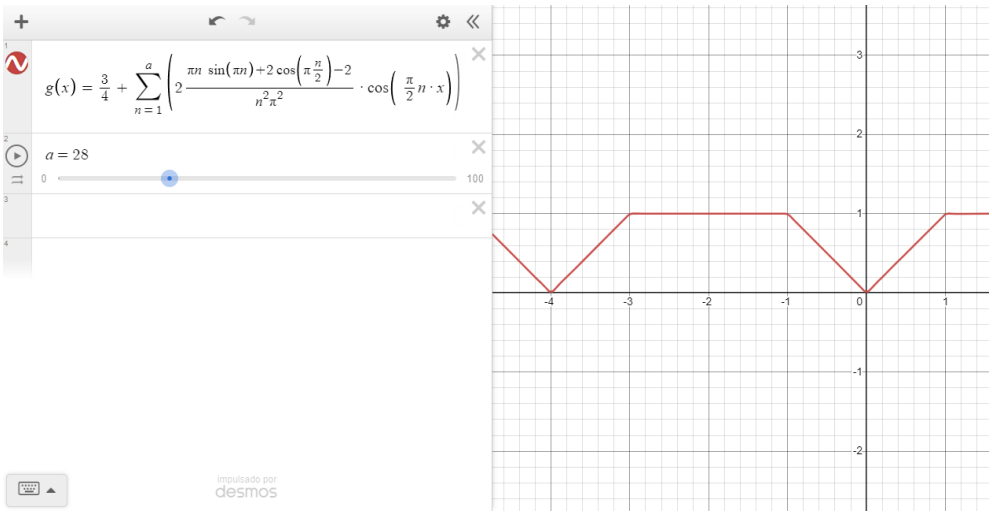
$4\left(\frac{u\,\text{sen}(\pi n u)}{\pi n} + \frac{\cos(\pi n u)}{\pi^2 n^2}\right)$

$\frac{2\left(\pi n x\,\text{sen}\left(\frac{\pi n x}{2}\right) + 2\cos\left(\frac{\pi n x}{2}\right)\right)}{\pi^2 n^2}\Bigg|_0^1 \Rightarrow \frac{2\left(\pi n\,\text{sen}\left(\frac{\pi n}{2}\right) + 2\cos\left(\frac{\pi n}{2}\right) - 2\right)}{\pi^2 n^2}$

$\int_1^2 \cos\left(n\frac{\pi}{2}t\right)dt \Rightarrow \frac{2}{\pi n}\,\text{sen}\left(\pi\frac{\pi}{2}t\right)\Big|_1^2$

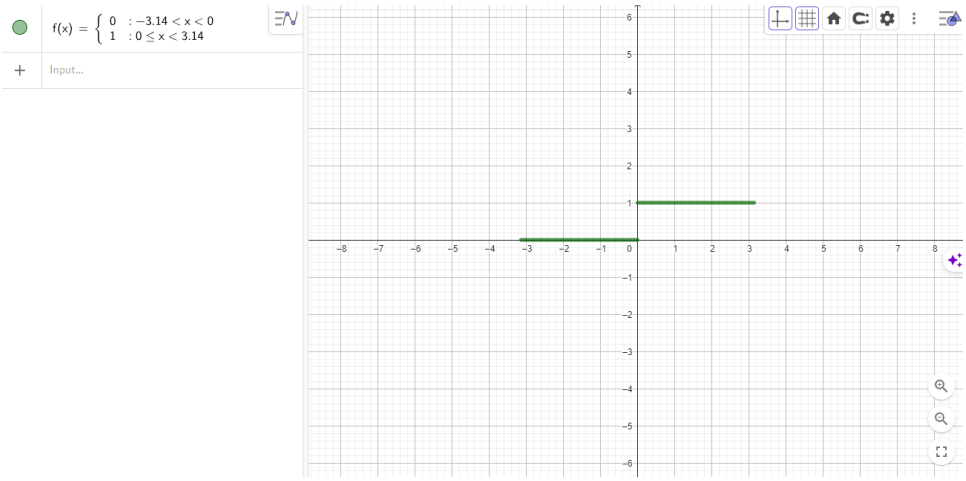
$\Downarrow$

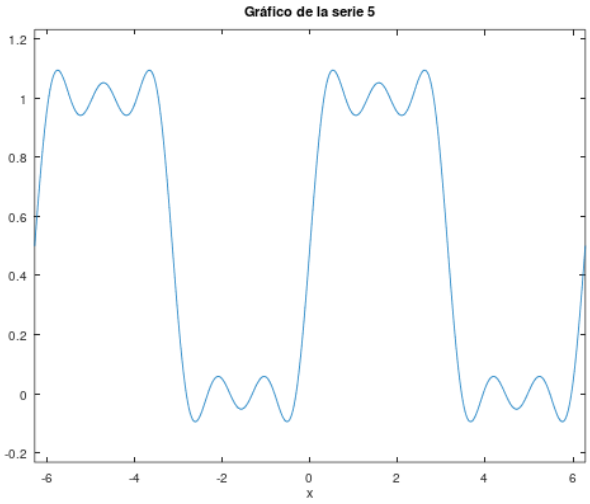
$\frac{2\left(\text{sen}(\pi n) - \text{sen}\left(\pi\frac{n}{2}\right)\right)}{\pi n}$

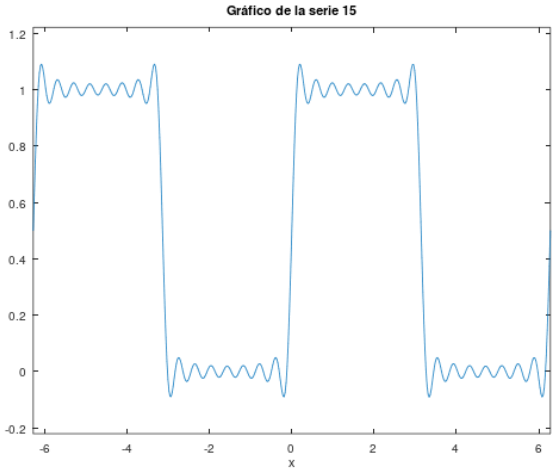


B) Para cada una de las funciones de los puntos 1, 3 y 4 deberá graficar la función de origen y la transformación asociada a la serie de Fourier. Para esto debe crear una función que resuelva la serie con diferentes N. Para el desarrollo de la función puede utilizar Octave o el software que considere adecuado. Deberá presentar el código y el ejecutable.

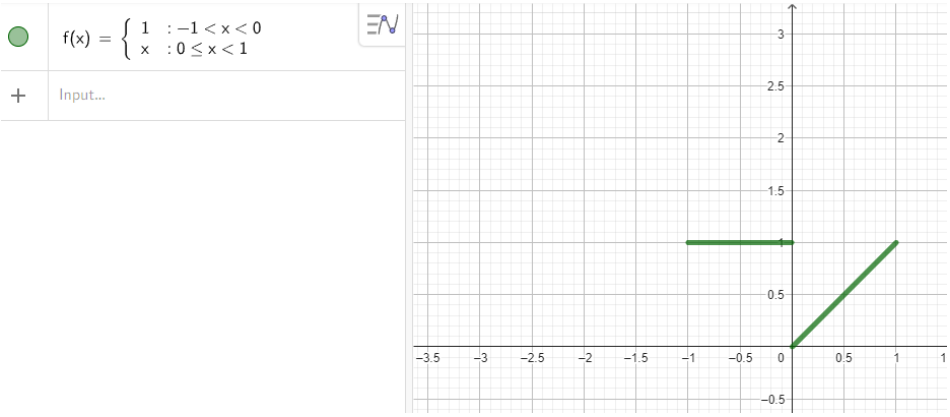
1) a)







b)


f(x) = { 1 : −1 < x < 0 ; x : 0 ≤ x < 1 }
Input...

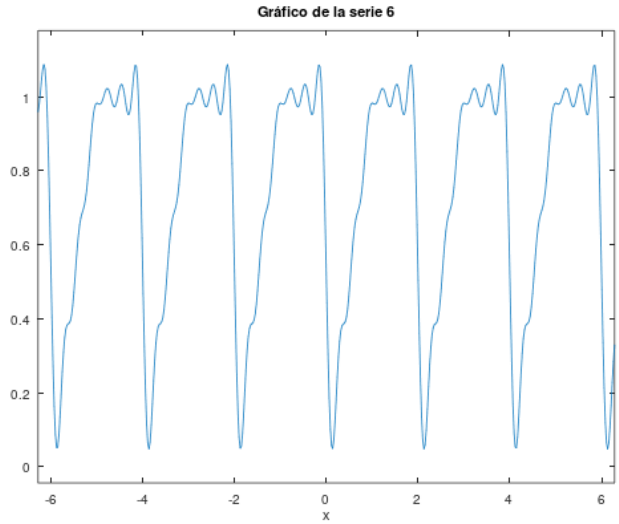


Gráfico de la serie 6
x

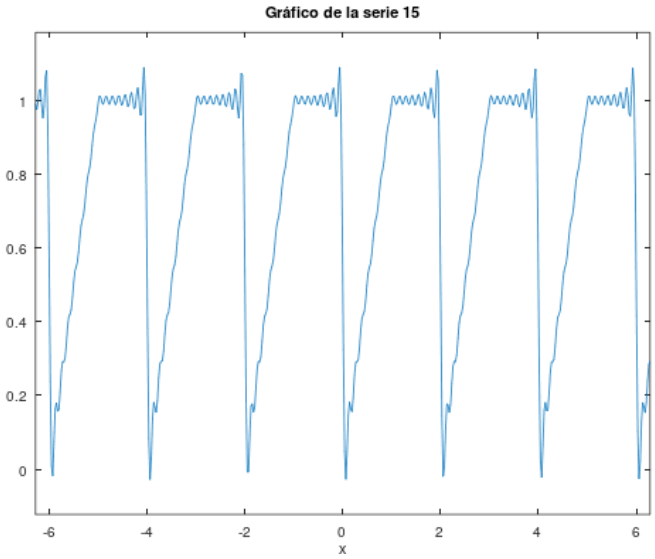
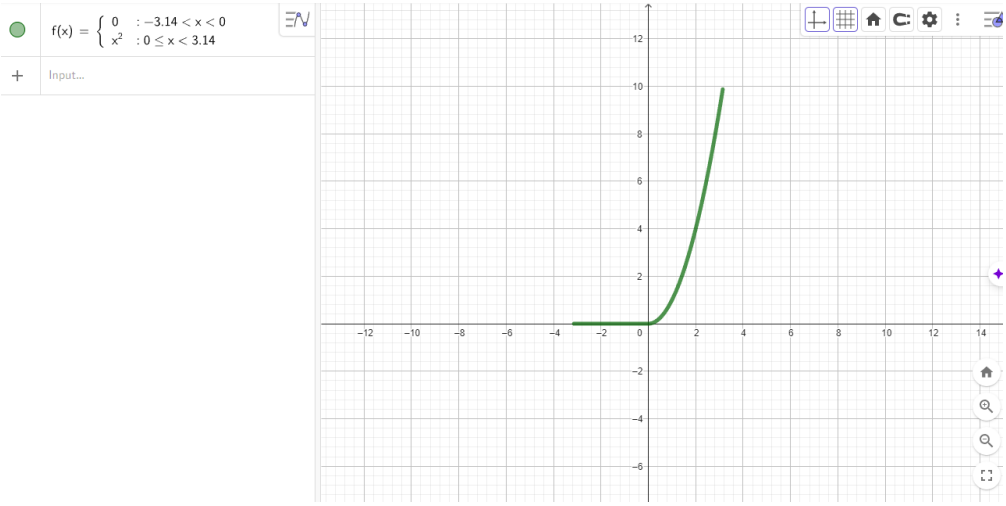


Gráfico de la serie 15
x

c)


f(x) = { 0 : −3.14 < x < 0 ; x² : 0 ≤ x < 3.14
Input...

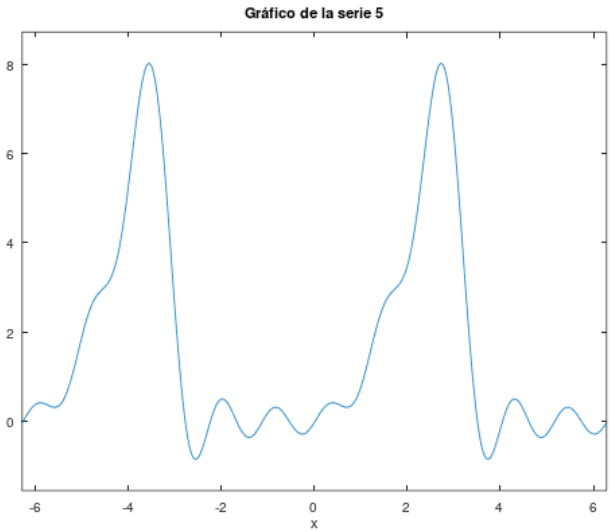


Gráfico de la serie 5
x

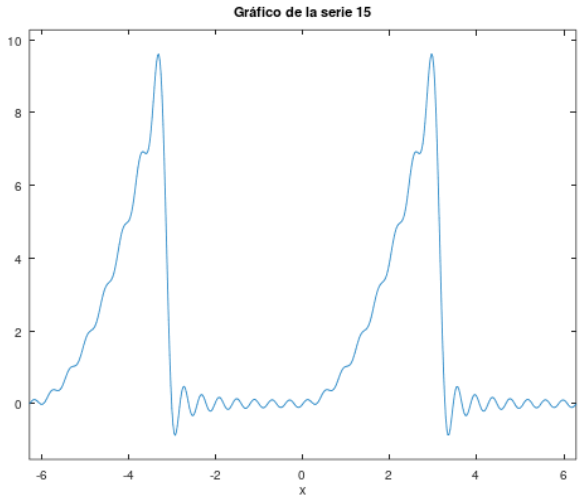
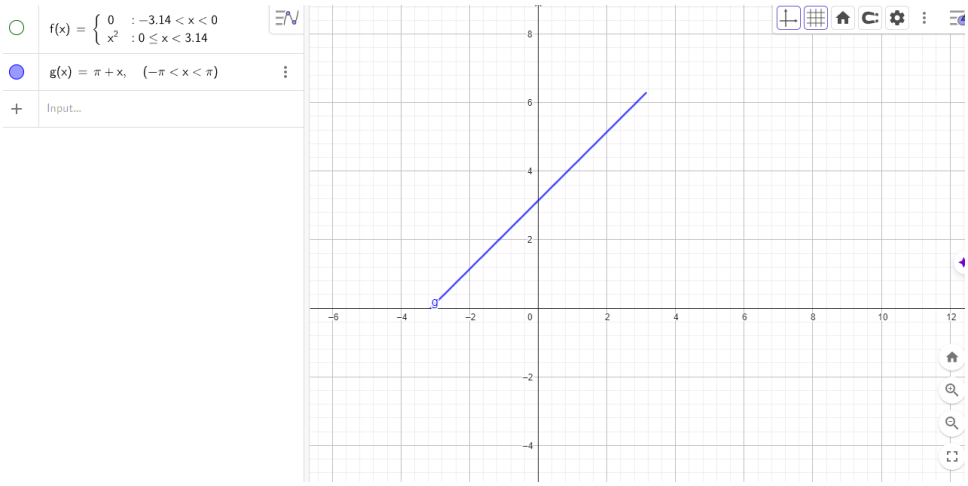


Gráfico de la serie 15
x

d)


f(x) = { 0 : −3.14 < x < 0 ; x² : 0 ≤ x < 3.14
g(x) = π + x, (−π < x < π)
Input...

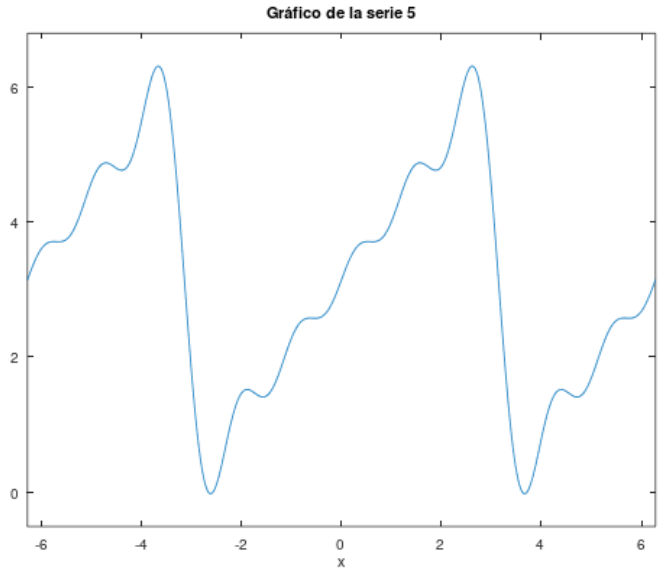


Gráfico de la serie 5
x

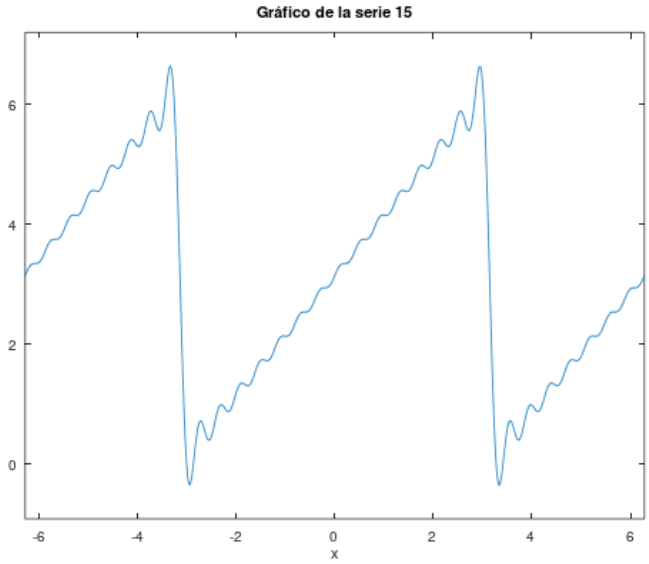
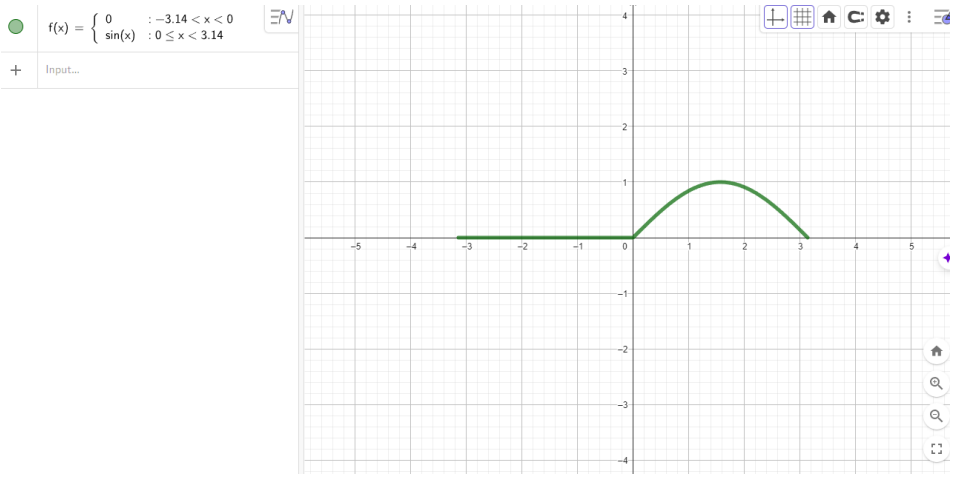


Gráfico de la serie 15
x

e)


f(x) = { 0 : −3.14 < x < 0 ; sin(x) : 0 ≤ x < 3.14
Input...

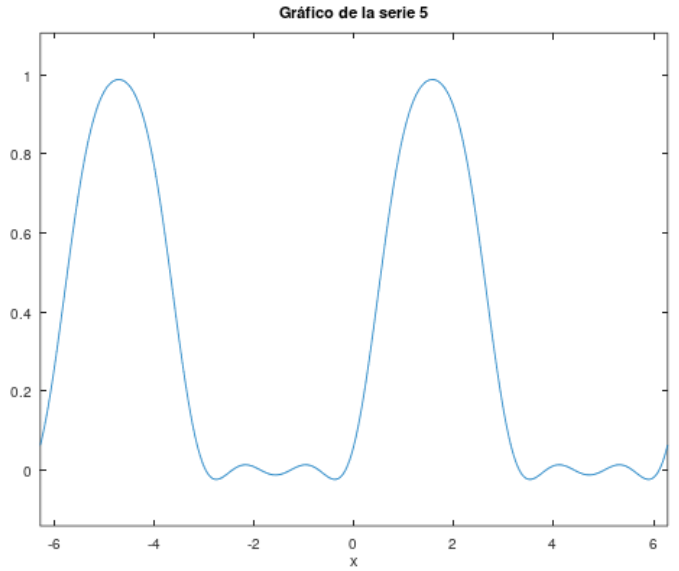


Gráfico de la serie 5
x

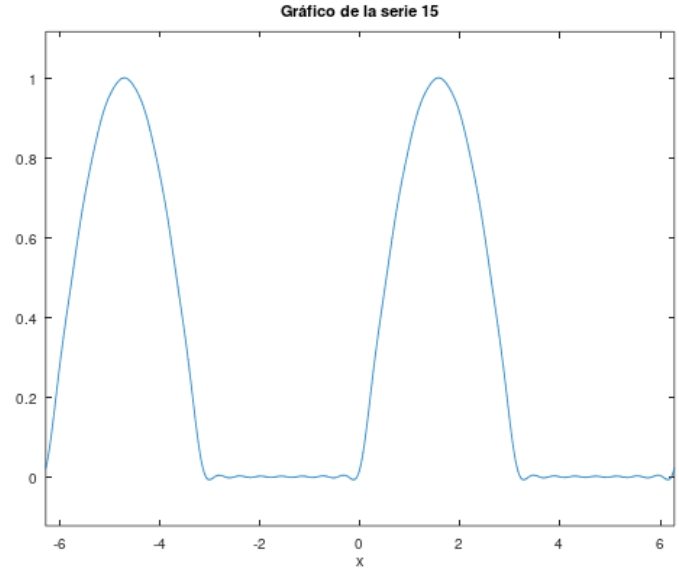
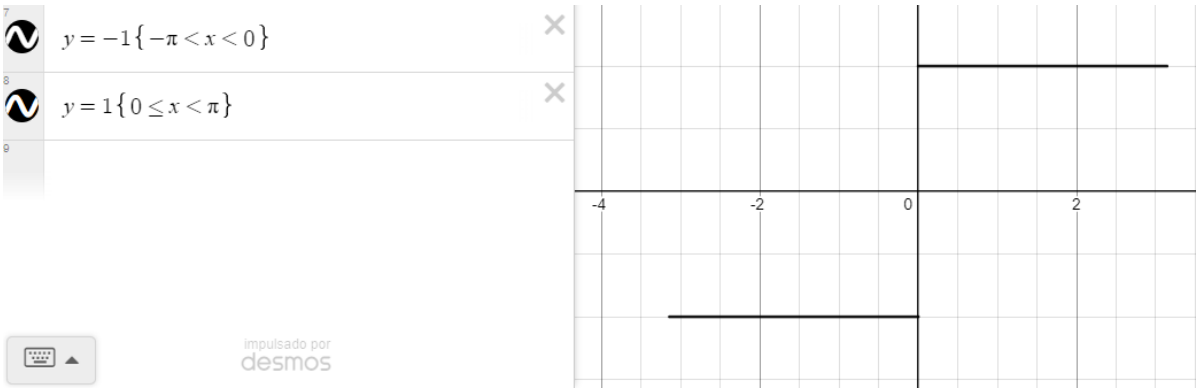


Gráfico de la serie 15
x

3)

a)


$y = -1\{-\pi < x < 0\}$
$y = 1\{0 \leq x < \pi\}$
-4
-2
0
2
impulsado por
desmos

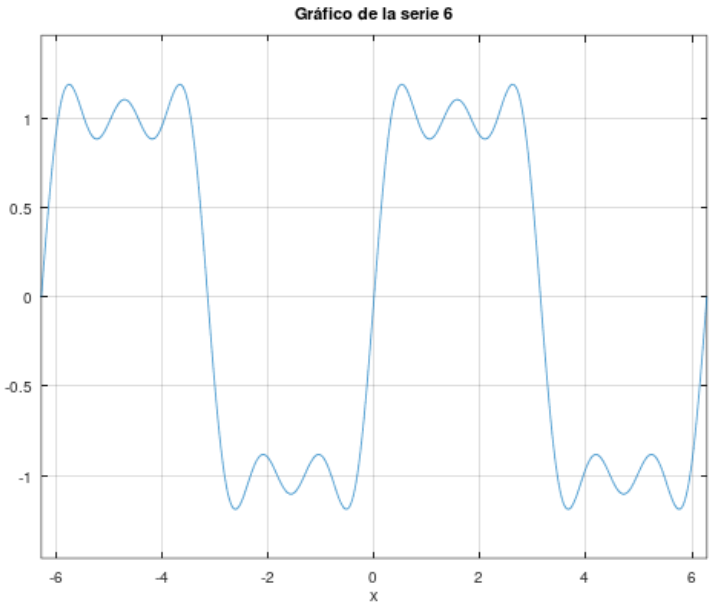


Gráfico de la serie 6
1
0.5
0
-0.5
-1
-6
-4
-2
0
2
4
6
x

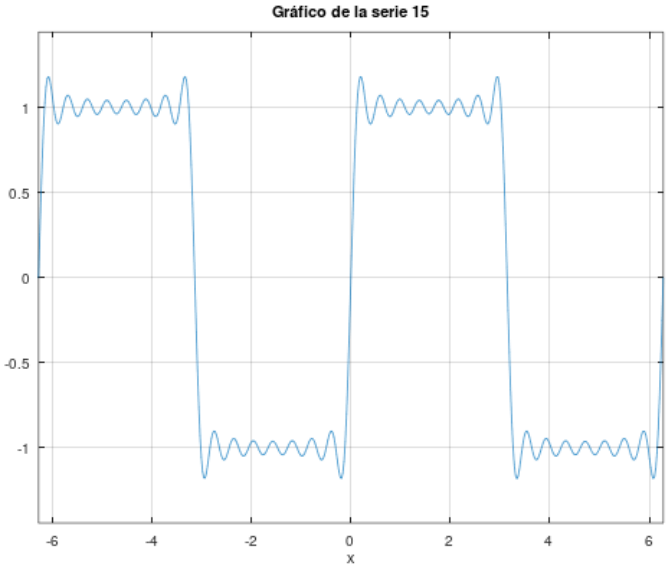
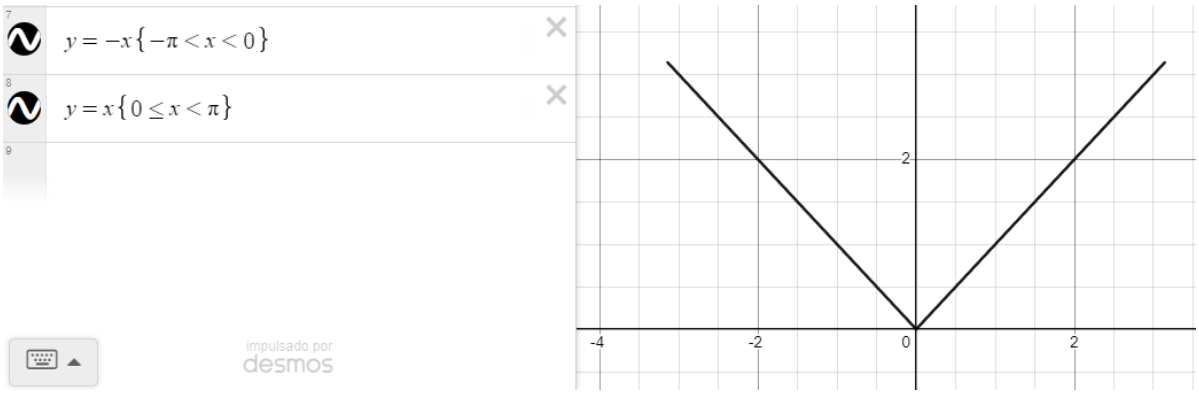


Gráfico de la serie 15
1
0.5
0
-0.5
-1
-6
-4
-2
0
2
4
6
x

b)


7
$y = -x\{-\pi < x < 0\}$
8
$y = x\{0 \leq x < \pi\}$
9
impulsado por
desmos
2
-4
-2
0
2

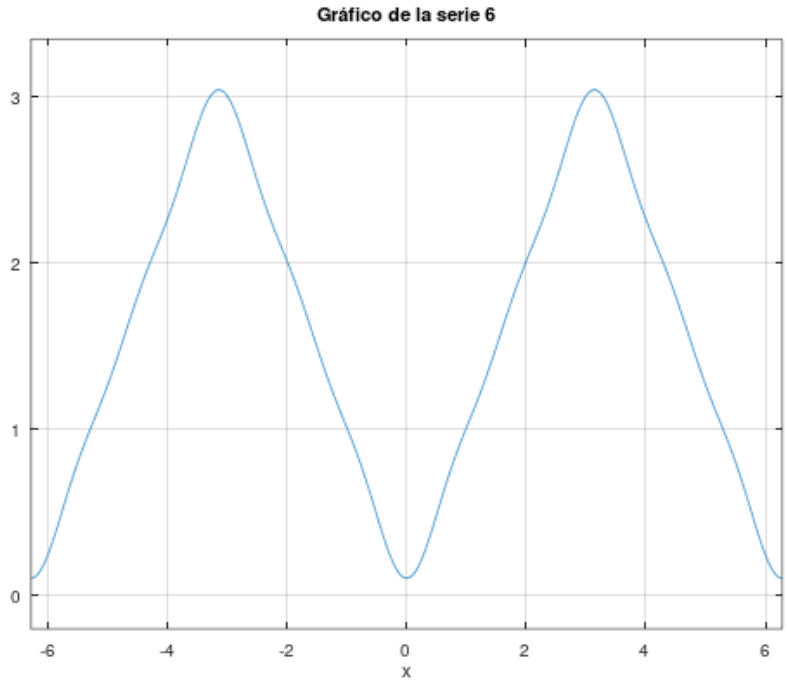


Gráfico de la serie 6
3
2
1
0
-6
-4
-2
0
2
4
6
x

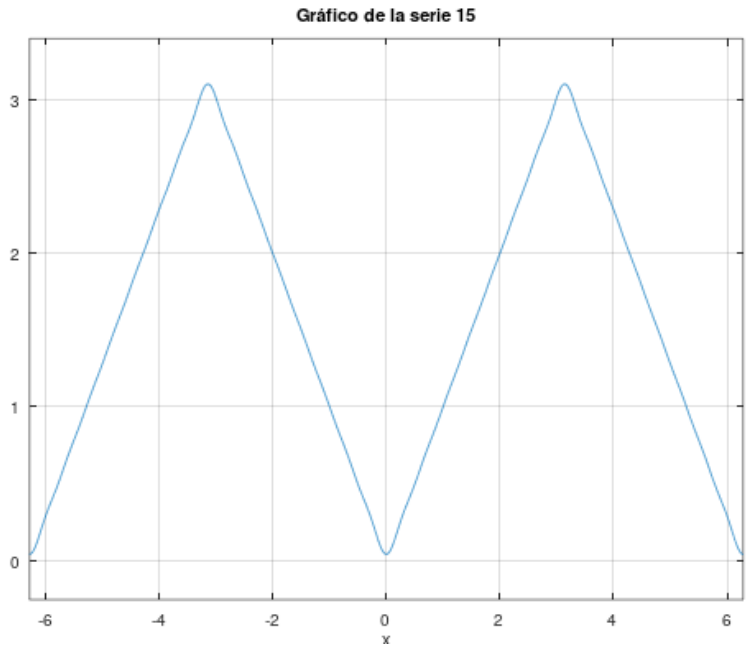
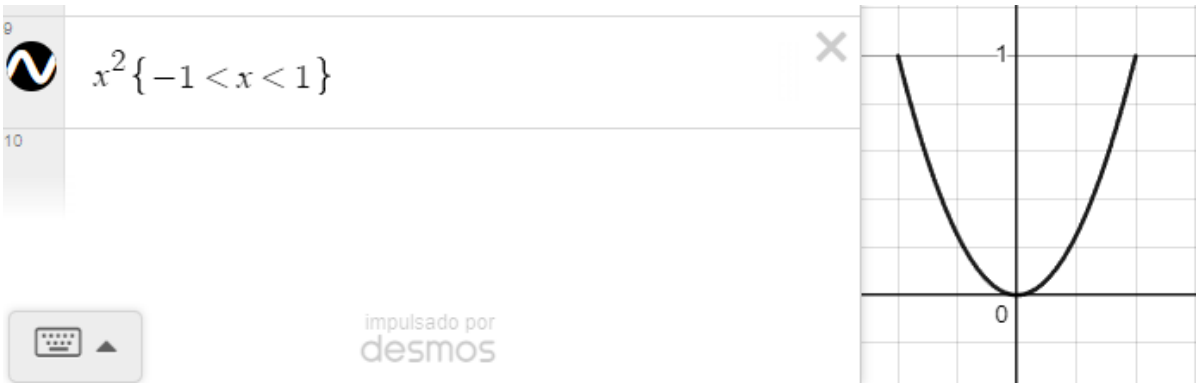


Gráfico de la serie 15
3
2
1
0
-6
-4
-2
0
2
4
6
x

c)


9
$x^2\{-1<x<1\}$
10
impulsado por
desmos
1
0

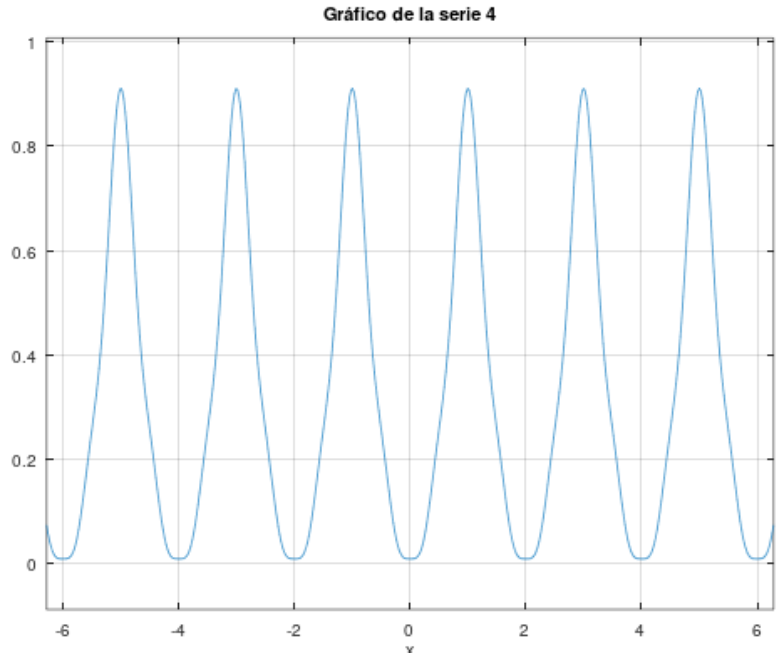


Gráfico de la serie 4
1
0.8
0.6
0.4
0.2
0
-6
-4
-2
0
2
4
6
x

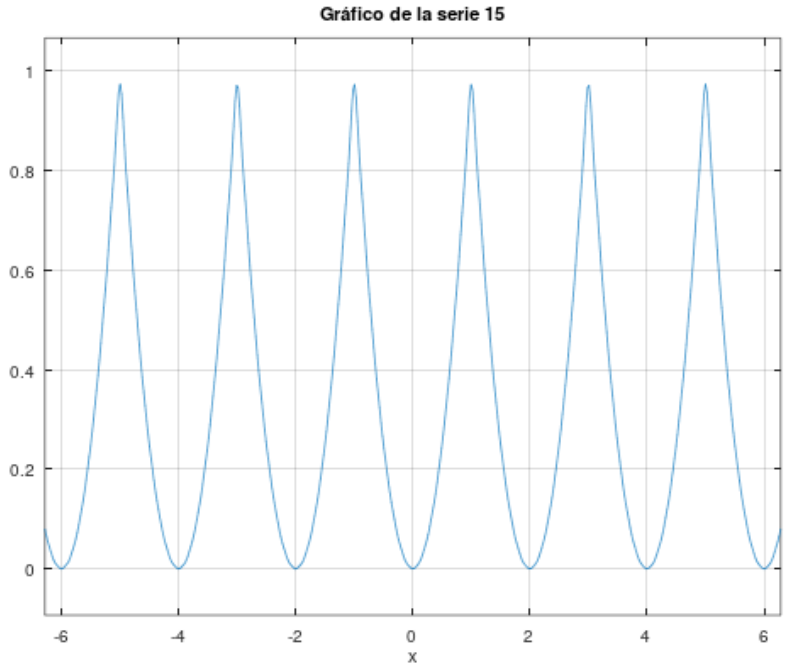
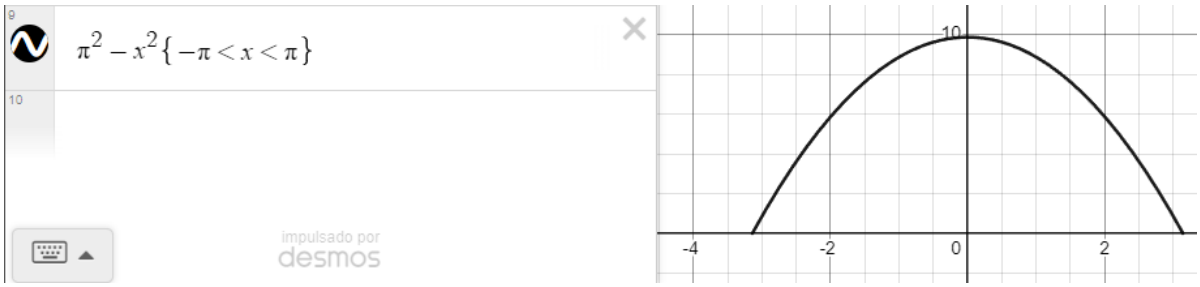


Gráfico de la serie 15
1
0.8
0.6
0.4
0.2
0
-6
-4
-2
0
2
4
6
x

d)


$\pi^2 - x^2\{-\pi < x < \pi\}$
impulsado por desmos
10
-4
-2
0
2

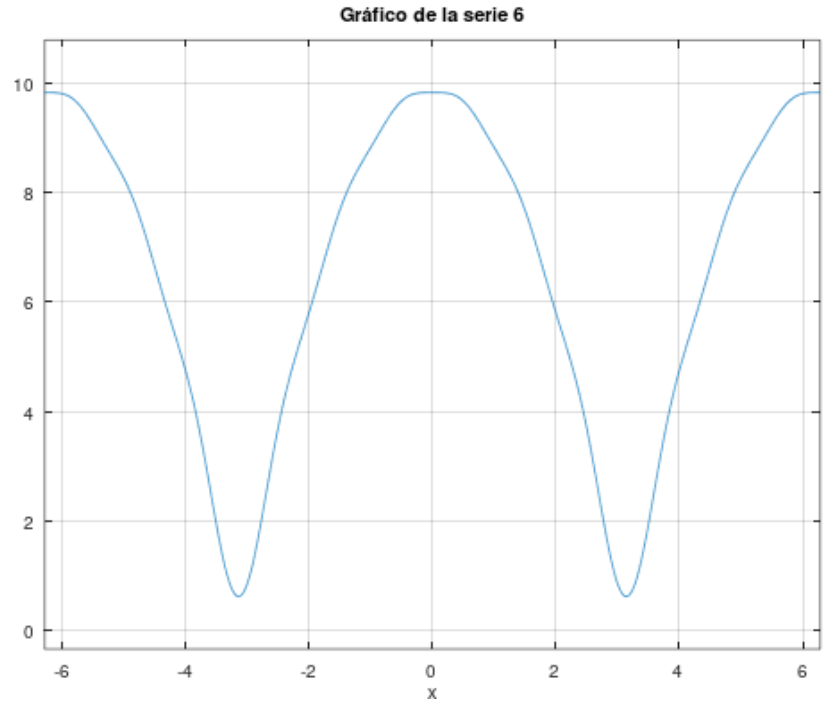


Gráfico de la serie 6
10
8
6
4
2
0
-6
-4
-2
0
2
4
6
x

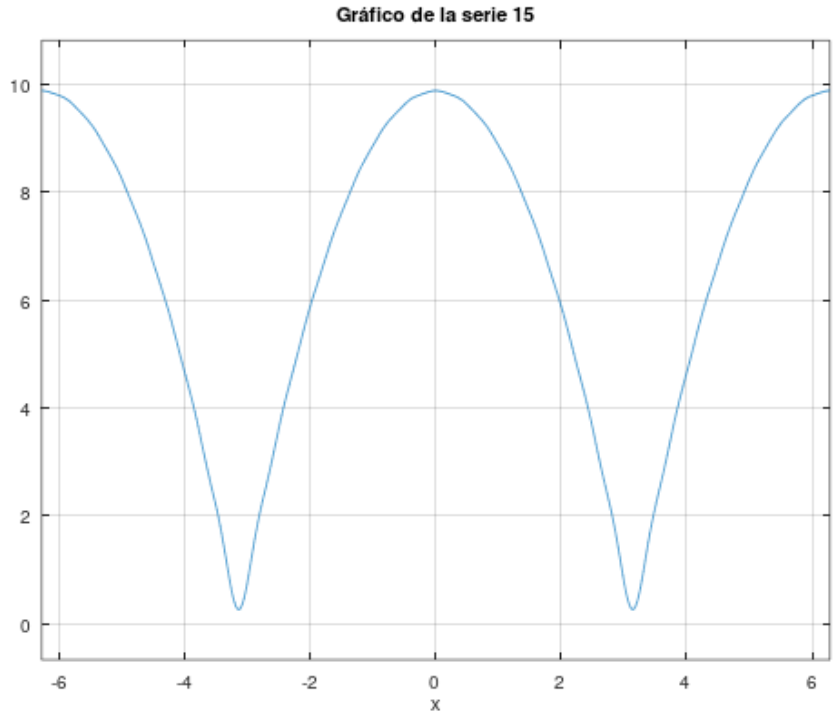
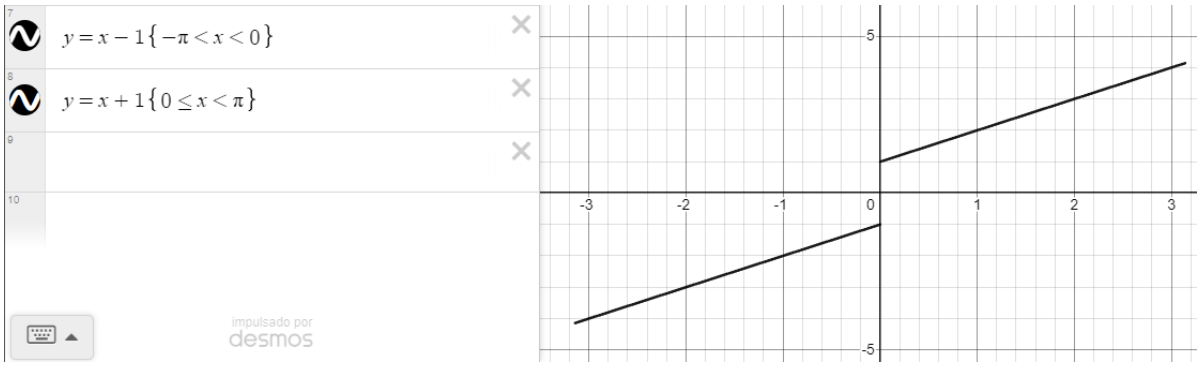


Gráfico de la serie 15
10
8
6
4
2
0
-6
-4
-2
0
2
4
6
x

e)


$y = x - 1\{-\pi < x < 0\}$
$y = x + 1\{0 \leq x < \pi\}$
impulsado por desmos

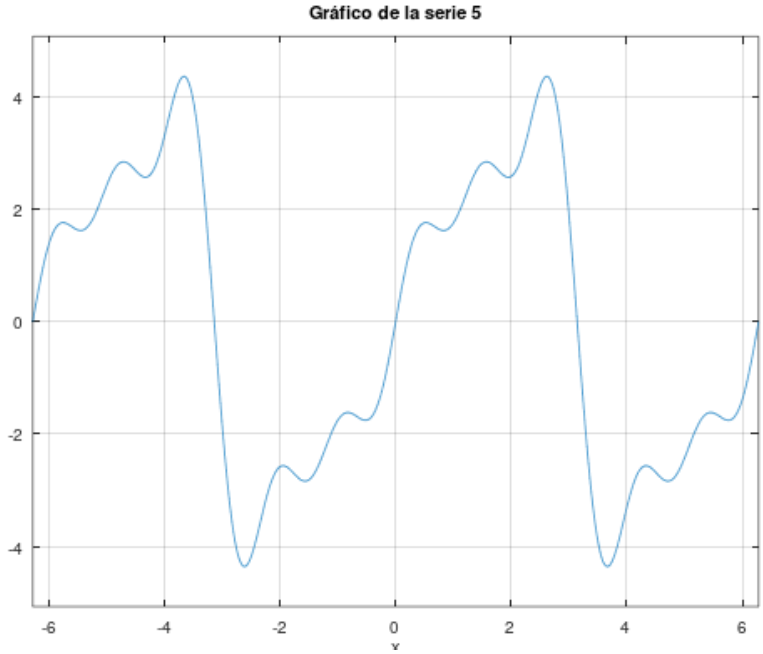


Gráfico de la serie 5

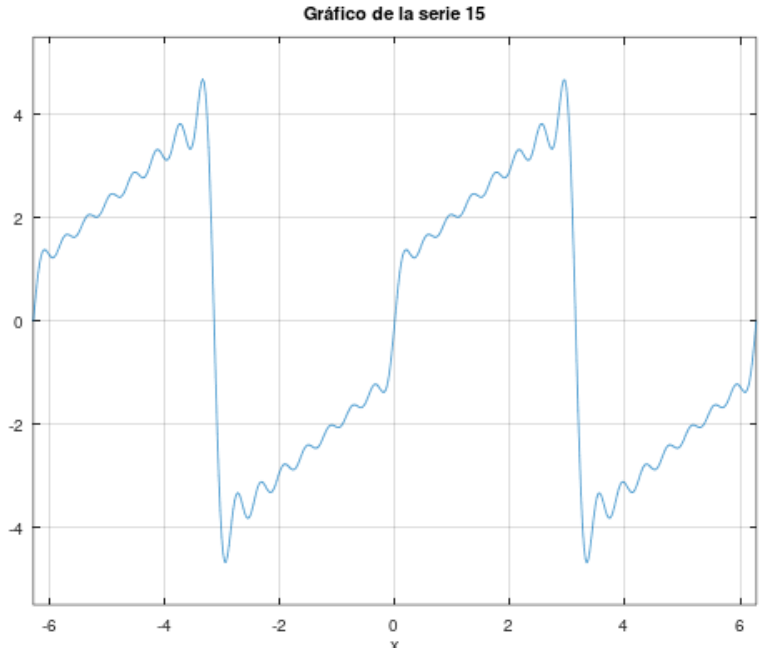
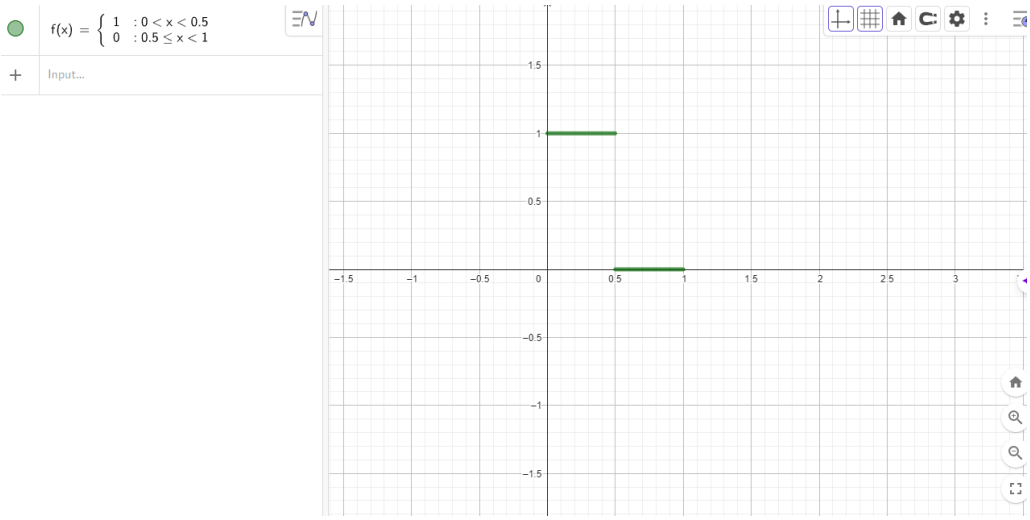


Gráfico de la serie 15

4) a)


f(x) = { 1 : 0 < x < 0.5 ; 0 : 0.5 ≤ x < 1 }
Input...

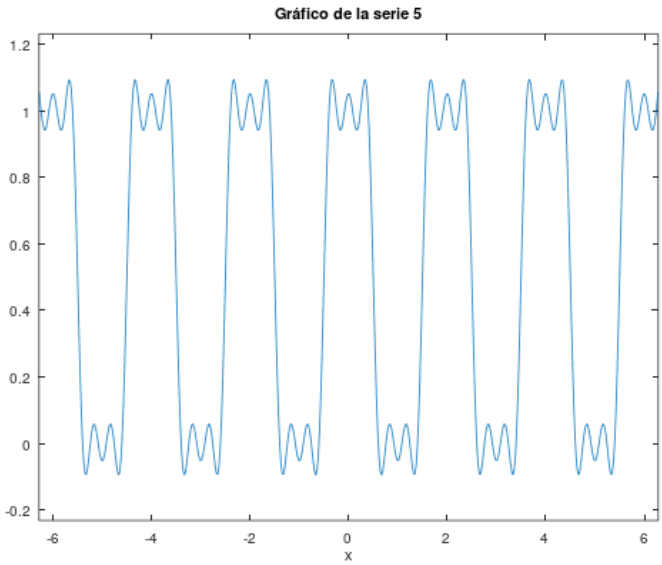
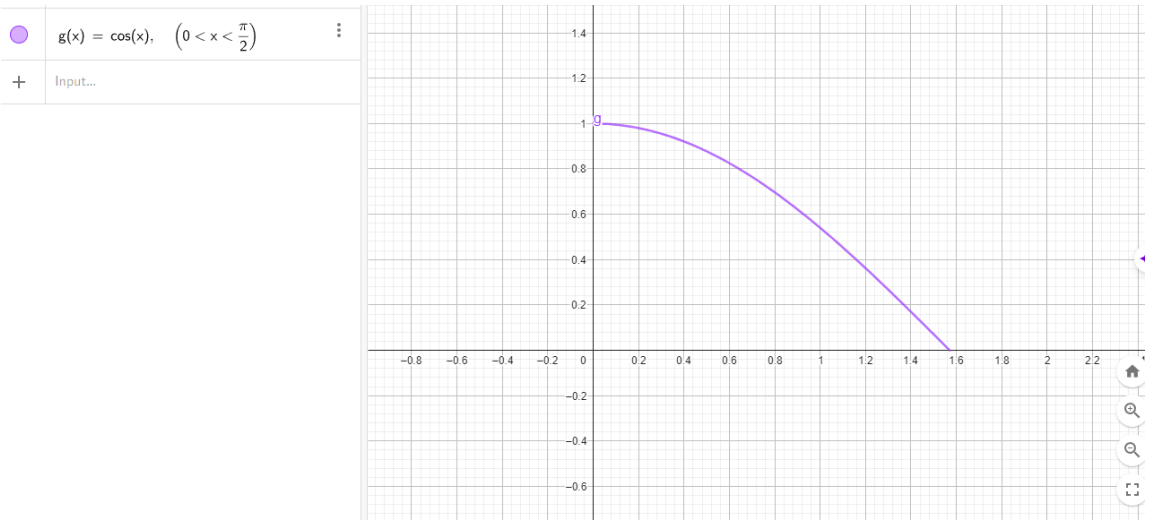


Gráfico de la serie 5
x

Gráfico de la serie 15
x

b)


g(x) = cos(x), (0 < x < π/2)
Input...

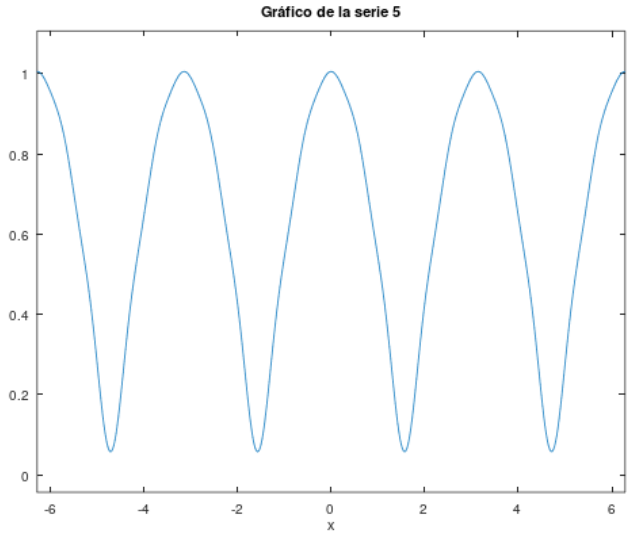


Gráfico de la serie 5
x

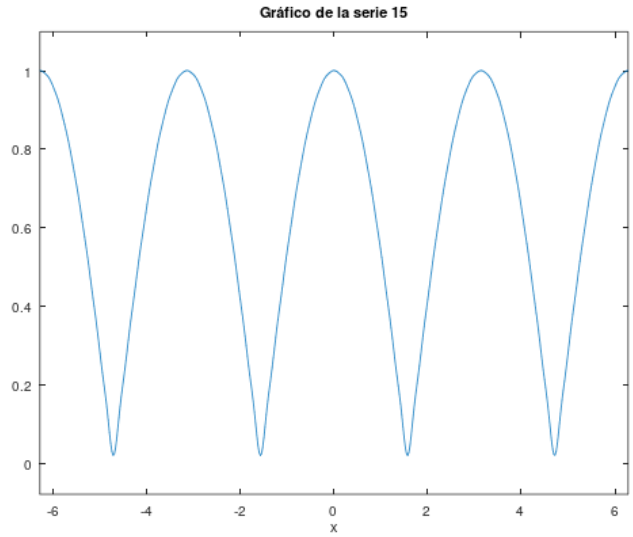
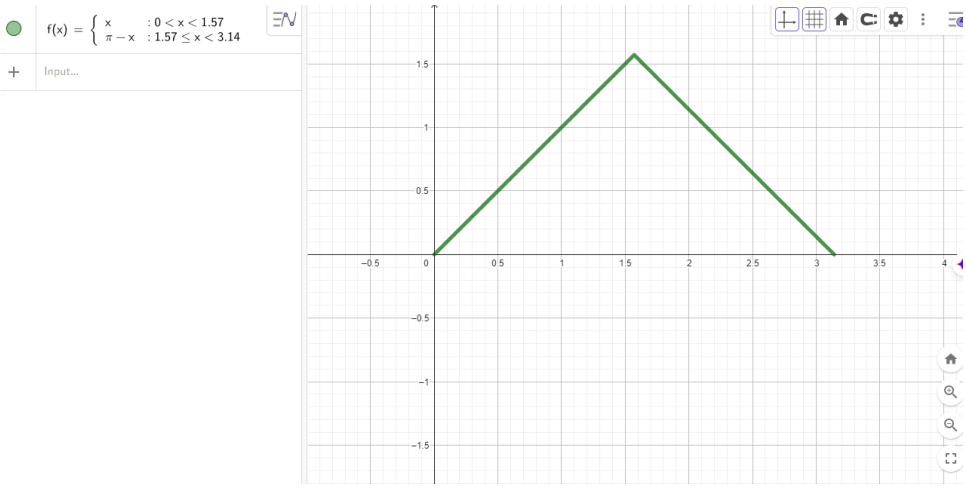


Gráfico de la serie 15
x

c)


f(x) = { x : 0 < x < 1.57 ; π − x : 1.57 ≤ x < 3.14
Input...

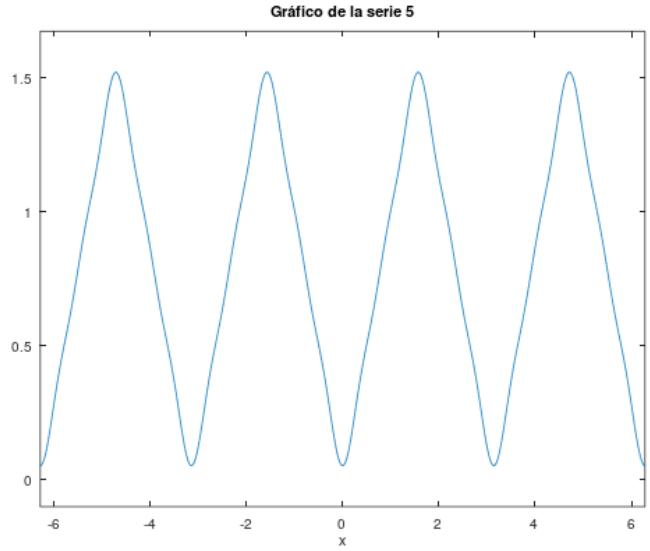


Gráfico de la serie 5
x

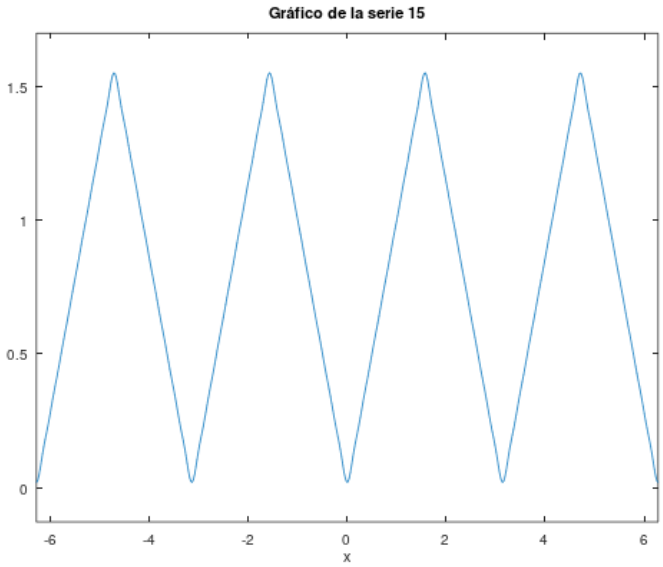
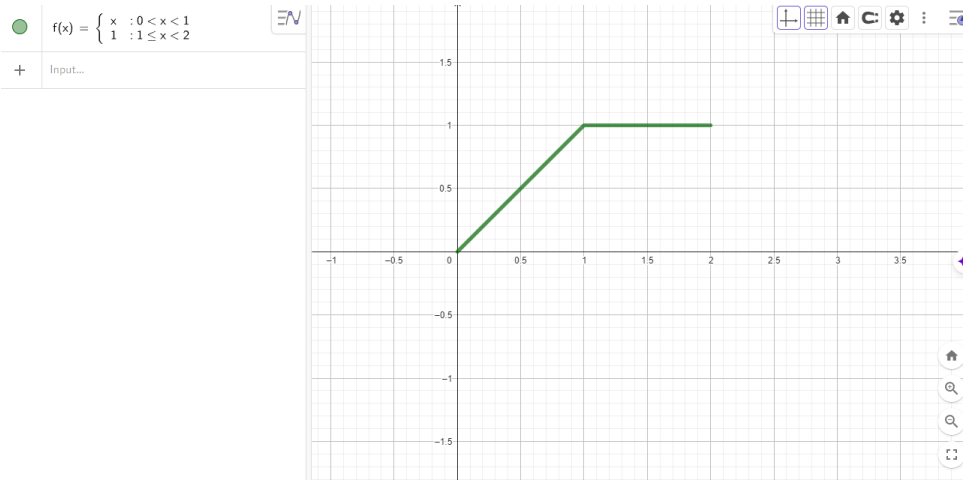


Gráfico de la serie 15
x

d)


f(x) = { x : 0 < x < 1 ; 1 : 1 ≤ x < 2
Input...

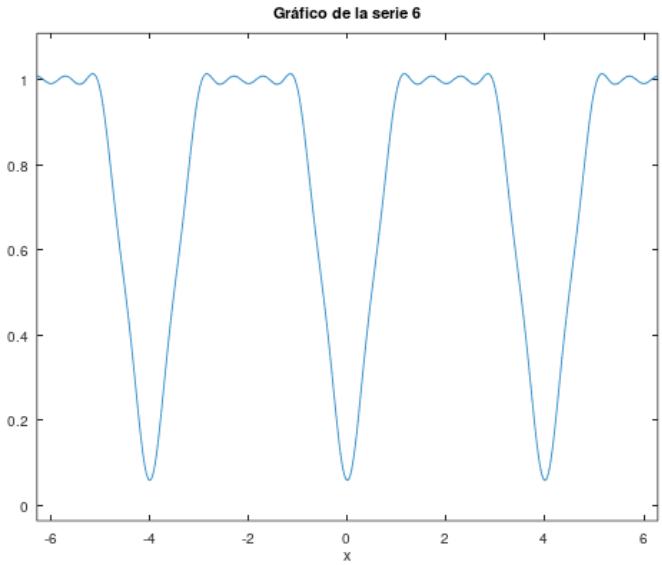


Gráfico de la serie 6
x

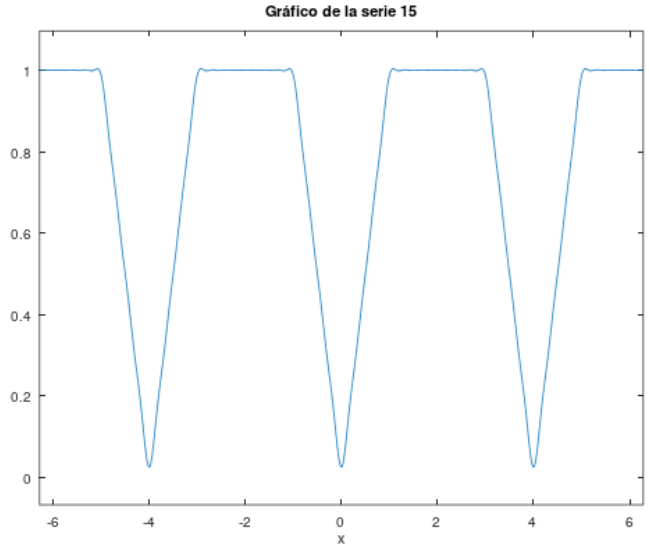


Gráfico de la serie 15
x

2) Resolver las transformadas de Fourier del punto 5

1- $f(t) = \begin{matrix} 1\ si\ |t| > 3 \\ 0\ si\ |t| < 3 \end{matrix} \Rightarrow f(t) = \begin{matrix} 1\ si\ \ t > 3\ \wedge\ t < -3 \\ 0\ si \qquad -3 < t < 3 \end{matrix}$

$$F(\omega) = \mathcal{F}[(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$F(\omega) = \int_{-\infty}^{-3} 1.e^{-j\omega t} dt + \int_{-3}^{3} 0.e^{-j\omega t} dt + \int_{3}^{\infty} 1.e^{-j\omega t} dt$$

$$F(\omega) = \lim_{a\to-\infty} \int_{a}^{-3} e^{-j\omega t} dt + \lim_{b\to\infty} \int_{3}^{b} e^{-j\omega t} dt$$

$$F(\omega) = \lim_{a\to-\infty} -\frac{e^{-j\omega t}}{j\omega}\Bigg]_{a}^{-3} + \lim_{b\to\infty} -\frac{e^{-j\omega t}}{j\omega}\Bigg]_{3}^{b}$$

$$F(\omega) = \lim_{a\to-\infty} \left[-\frac{e^{3j\omega}}{j\omega} + \frac{e^{aj\omega}}{j\omega}\right] + \lim_{b\to\infty} \left[-\frac{e^{-bj\omega}}{j\omega} + \frac{e^{-3j\omega}}{j\omega}\right]$$

$$F(\omega) = -\frac{e^{3j\omega}}{j\omega} + \lim_{a\to-\infty} \left[\frac{e^{aj\omega}}{j\omega}\right] + \frac{e^{-3j\omega}}{j\omega} + \lim_{b\to\infty} \left[-\frac{e^{-bj\omega}}{j\omega}\right]$$

$$\lim_{x\to-\infty} e^{x} = 0, \lim_{x\to\infty} e^{-x} = 0$$

$$F(\omega) = -\frac{e^{3j\omega}}{j\omega} + \frac{e^{-3j\omega}}{j\omega} = \frac{e^{-3j\omega}}{j\omega} - \frac{e^{3j\omega}}{j\omega} = \frac{e^{-3j\omega} - e^{3j\omega}}{j\omega} = \frac{2}{\omega}\frac{e^{-3j\omega} - e^{3j\omega}}{j2}$$

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{i2}$$

$$F(\omega) = \frac{2}{\omega}\sin(-3\omega)$$

2- $f(t) = \begin{matrix} e^{-t}\ si\ t > 0 \\ 0\ si\ t < 0 \end{matrix}$

$$F(\omega) = \mathcal{F}[(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$F(\omega) = \int_{-\infty}^{0} 0.e^{-j\omega t} dt + \int_{0}^{\infty} e^{-t}.e^{-j\omega t} dt$$

$$F(\omega) = \int_{0}^{\infty} e^{-j\omega t - t} dt = \int_{0}^{\infty} e^{(-j\omega - 1)t} dt$$

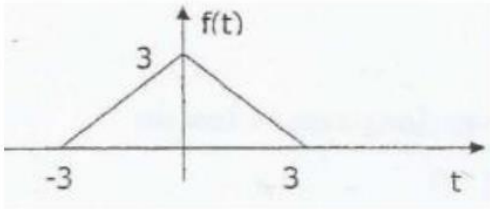
$$F(\omega) = \lim_{a\to\infty} \int_{0}^{a} e^{(-j\omega-1)t} dt$$

$$F(\omega) = \lim_{a\to\infty} \frac{e^{(-j\omega-1)t}}{-j\omega - 1}\Bigg]_{0}^{a} = \lim_{a\to\infty} \left[\frac{e^{(-j\omega-1)a}}{-j\omega - 1} - \frac{e^{(-j\omega-1)0}}{-j\omega - 1}\right]$$

$$F(\omega) = -\frac{1}{-j\omega - 1} + \lim_{a\to\infty} \left[\frac{e^{-a(j\omega+1)}}{-j\omega - 1}\right]$$

$$\lim_{x\to\infty} e^{-x} = 0$$

$$F(\omega) = -\frac{1}{-j\omega - 1} = \frac{1}{j\omega + 1} * \frac{j\omega - 1}{j\omega - 1} = \frac{j\omega - 1}{(j\omega)^2 - 1^2} = \frac{j\omega - 1}{-\omega^2 - 1^2} = \frac{-j\omega + 1}{\omega^2 + 1}$$

3- 

$$f(t) = \begin{matrix} 3+t \;\; si -3 < t < 0 \\ 3-t \quad si\; 0 \leq t < 3 \end{matrix}$$

$$F(\omega) = \mathcal{F}[(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

$$F(\omega) = \int_{-\infty}^{-3} 0.e^{-j\omega t}dt + \int_{-3}^{0}(3+t)e^{-j\omega t}dt + \int_{0}^{3}(3-t)e^{-j\omega t}dt + \int_{3}^{\infty} 0.e^{-j\omega t}dt$$

$$F(\omega) = \int_{-3}^{0} 3e^{-j\omega t}dt + \int_{-3}^{0} t.e^{-j\omega t}dt + \int_{0}^{3} 3e^{-j\omega t}dt - \int_{0}^{3} t.e^{-j\omega t}dt$$

$$a - \int_{-3}^{0} 3e^{-j\omega t}dt = \frac{3e^{-j\omega t}}{-j\omega}\Bigg]_{-3}^{0} = \frac{3e^{-j\omega 0}}{-j\omega} - \frac{3e^{3j\omega}}{-j\omega} = \frac{3}{-j\omega} - \frac{3e^{3j\omega}}{-j\omega} = \frac{3-3e^{3j\omega}}{-j\omega}$$

$$b - \int_{-3}^{0} t.e^{-j\omega t}dt \begin{bmatrix} u = t \rightarrow du = 1dt \\ dv = e^{-j\omega t}dt \rightarrow v = \dfrac{e^{-j\omega t}}{-j\omega} \end{bmatrix} por \int_a^b udv = uv]_a^b - \int_a^b vdu$$

$$\int_{-3}^{0} t.e^{-j\omega t}dt = x.\frac{e^{-j\omega t}}{-j\omega}\Bigg]_{-3}^{0} - \int_{-3}^{0}\frac{e^{-j\omega t}}{-j\omega}.1dt = \frac{3e^{3j\omega}}{-j\omega} - \frac{e^{-j\omega t}}{j^2\omega^2}\Bigg]_{-3}^{0}$$

$$\int_{-3}^{0} t.e^{-j\omega t}dt = \frac{3e^{3j\omega}}{-j\omega} - \frac{1}{j^2\omega^2} - \frac{e^{3j\omega}}{j^2\omega^2} = \frac{3e^{3j\omega}}{-j\omega} - \frac{1-e^{3j\omega}}{j^2\omega^2}$$

$$c - \int_{0}^{3} 3e^{-j\omega t}dt = \frac{3e^{-j\omega t}}{-j\omega}\Bigg]_{0}^{3} = \frac{3e^{-3j\omega}}{-j\omega} - \frac{3e^{-j\omega 0}}{-j\omega} = \frac{3e^{-3j\omega}}{-j\omega} - \frac{3}{-j\omega} = \frac{3e^{3j\omega}-3}{-j\omega}$$

$$d - \int_{0}^{3} t.e^{-j\omega t}dt \begin{bmatrix} u = t \rightarrow du = 1dt \\ dv = e^{-j\omega t}dt \rightarrow v = \dfrac{e^{-j\omega t}}{-j\omega} \end{bmatrix} por \int_a^b udv = uv]_a^b - \int_a^b vdu$$

$$\int_{0}^{3} t.e^{-j\omega t}dt = x.\frac{e^{-j\omega t}}{-j\omega}\Bigg]_{0}^{3} - \int_{0}^{3}\frac{e^{-j\omega t}}{-j\omega}.1dt = \frac{3e^{-3j\omega}}{-j\omega} - \frac{e^{-j\omega t}}{j^2\omega^2}\Bigg]_{0}^{3}$$

$$\int_{0}^{3} t.e^{-j\omega t}dt = \frac{3e^{3j\omega}}{-j\omega} - \frac{e^{-3j\omega}}{j^2\omega^2} - \frac{1}{j^2\omega^2} = \frac{3e^{3j\omega}}{-j\omega} - \frac{e^{-3j\omega}-1}{j^2\omega^2}$$

$$F(\omega) = \frac{3-3e^{3j\omega}}{-j\omega} + \frac{3e^{3j\omega}}{-j\omega} - \frac{1-e^{3j\omega}}{j^2\omega^2} + \frac{3e^{3j\omega}-3}{-j\omega} - \left(\frac{3e^{3j\omega}}{-j\omega} - \frac{e^{-3j\omega}-1}{j^2\omega^2}\right)$$

$$F(\omega) = \frac{3-3e^{3j\omega}}{-j\omega} + \frac{3e^{3j\omega}}{-j\omega} - \frac{1-e^{3j\omega}}{j^2\omega^2} + \frac{3e^{3j\omega}-3}{-j\omega} - \frac{3e^{3j\omega}}{-j\omega} + \frac{e^{-3j\omega}-1}{j^2\omega^2}$$

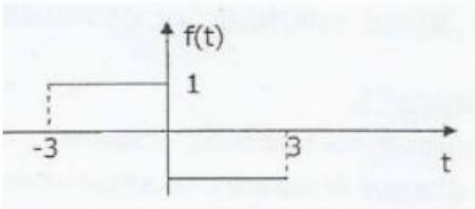
$$F(\omega) = \frac{3-3e^{3j\omega}+3e^{3j\omega}-3}{-j\omega} - \frac{1-e^{3j\omega}+e^{-3j\omega}-1}{j^2\omega^2}$$

$$F(\omega) = -\frac{e^{3j\omega}+e^{-3j\omega}}{j^2\omega^2}$$

$$F(\omega) = -\frac{e^{3j\omega}+e^{-3j\omega}}{j^2\omega^2}$$

$$\cos(z) = \frac{e^{iz}+e^{-iz}}{2}$$

$$F(\omega) = -\frac{2}{j^2\omega^2}\frac{e^{3j\omega}+e^{-3j\omega}}{2} = \frac{2}{\omega^2}\cos(3\omega)$$

4- 

$$f(t) = \begin{matrix} 1 & si -3 < t < 0 \\ -1 & si\ 0 \le t < 3 \end{matrix}$$

$$F(\omega) = \mathcal{F}[(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

$$F(\omega) = \int_{-\infty}^{-3} 0.e^{-j\omega t}dt + \int_{-3}^{0} 1.e^{-j\omega t}dt + \int_{0}^{3} -1.e^{-j\omega t}dt + \int_{3}^{\infty} 0.e^{-j\omega t}dt$$

$$F(\omega) = \int_{-3}^{0} e^{-j\omega t}dt + \int_{0}^{3} -e^{-j\omega t}dt$$

$$F(\omega) = \left.\frac{e^{-j\omega t}}{-j\omega}\right]_{-3}^{0} + \left.\frac{e^{-j\omega t}}{j\omega}\right]_{0}^{3} = \frac{1}{-j\omega} - \frac{e^{3j\omega}}{-j\omega} + \frac{e^{-3j\omega}}{j\omega} - \frac{1}{j\omega}$$

$$= \frac{-1}{j\omega} - \frac{-e^{3j\omega}}{j\omega} + \frac{e^{-3j\omega}}{j\omega} - \frac{1}{j\omega}$$

$$F(\omega) = \frac{-1+e^{3j\omega}+e^{-3j\omega}-1}{j\omega} = \frac{e^{3j\omega}+e^{-3j\omega}-2}{j\omega} = \frac{e^{3j\omega}+e^{-3j\omega}}{j\omega} - \frac{2}{j\omega}$$

$$\cos(z) = \frac{e^{iz}+e^{-iz}}{2}$$

$$F(\omega) = \frac{2}{j\omega}\frac{e^{3j\omega}+e^{-3j\omega}}{2} - \frac{2}{j\omega} = \frac{2}{j\omega}\cos(3\omega) - \frac{2}{j\omega} = \frac{2}{j\omega}(\cos(3\omega)-1)$$

$$F(\omega) = \frac{-2j}{\omega}(\cos(3\omega)-1)$$

5- $f(t) = \begin{matrix} t & si\ 0 \le t \le 5 \\ 0 & si\ t < 0 \wedge t > 5 \end{matrix}$

$$F(\omega) = \mathcal{F}[(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

$$F(\omega) = \int_{-\infty}^{0} 0.e^{-j\omega t}dt + \int_{0}^{5} t.e^{-j\omega t}dt + \int_{5}^{\infty} 0.e^{-j\omega t}dt$$

$$F(\omega) = \int_{0}^{5} t.e^{-j\omega t}dt \begin{bmatrix} u = t \to du = 1dt \\ dv = e^{-j\omega t}dt \to v = \frac{e^{-j\omega t}}{-j\omega} \end{bmatrix} por \int_a^b udv = uv]_a^b - \int_a^b vdu$$

$$\int_0^5 t.e^{-j\omega t}dt = x.\left.\frac{e^{-j\omega t}}{-j\omega}\right]_0^5 - \int_0^5 \frac{e^{-j\omega t}}{-j\omega}.1dt = \left.\frac{5e^{-5j\omega}}{-j\omega} - \frac{e^{-j\omega t}}{j^2\omega^2}\right]_0^5$$

$$\int_0^5 t.e^{-j\omega t}dt = \frac{-5e^{-5j\omega}}{j\omega} - \frac{e^{-5j\omega}}{j^2\omega^2} - \frac{1}{j^2\omega^2} = \frac{-5e^{-5j\omega}}{j\omega} + \frac{e^{-5j\omega}}{\omega^2} + \frac{1}{\omega^2}$$

$$F(\omega) = \frac{5j}{\omega}e^{-5j\omega} + \frac{1}{\omega^2}e^{-5j\omega} + \frac{1}{\omega^2} = \left(\frac{1}{\omega^2} + \frac{5}{\omega}j\right)e^{-5j\omega} + \frac{1}{\omega^2}$$

6- $f(t) = \begin{matrix} 3-t & si\ 0 \le t \le 3 \\ 0 & si\ t < 0 \wedge t > 3 \end{matrix}$

$$F(\omega) = \mathcal{F}[(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

$$F(\omega) = \int_{-\infty}^{0} 0.e^{-j\omega t}dt + \int_{0}^{3} (3-t).e^{-j\omega t}dt + \int_{5}^{\infty} 0.e^{-j\omega t}dt$$

$$F(\omega) = \int_{0}^{3} 3e^{-j\omega t}dt - \int_{0}^{3} t.e^{-j\omega t}dt$$

$$a - \int_{0}^{3} 3e^{-j\omega t}dt = \left.\frac{3e^{-j\omega t}}{-j\omega}\right]_{0}^{3} = \frac{3e^{-3j\omega}}{-j\omega} - \frac{3e^{-j\omega 0}}{-j\omega} = \frac{3e^{-3j\omega}}{-j\omega} - \frac{3}{-j\omega} = \frac{3e^{3j\omega}-3}{-j\omega}$$

$$b - \int_{0}^{3} t.e^{-j\omega t}dt \left[\begin{matrix} u = t \rightarrow du = 1dt \\ dv = e^{-j\omega t}dt \rightarrow v = \dfrac{e^{-j\omega t}}{-j\omega} \end{matrix}\right] por \int_{a}^{b} udv = uv]_{a}^{b} - \int_{a}^{b} vdu$$

$$\int_{0}^{3} t.e^{-j\omega t}dt = x.\left.\frac{e^{-j\omega t}}{-j\omega}\right]_{0}^{3} - \int_{0}^{3} \frac{e^{-j\omega t}}{-j\omega}.1dt = \left.\frac{3e^{-3j\omega}}{-j\omega} - \frac{e^{-j\omega t}}{j^2\omega^2}\right]_{0}^{3}$$

$$\int_{0}^{3} t.e^{-j\omega t}dt = \frac{3e^{3j\omega}}{-j\omega} - \frac{e^{-3j\omega}}{j^2\omega^2} - \frac{1}{j^2\omega^2} = \frac{3e^{3j\omega}}{-j\omega} - \frac{e^{-3j\omega}-1}{j^2\omega^2}$$

$$F(\omega) = \frac{3e^{3j\omega}-3}{-j\omega} - \left(\frac{3e^{3j\omega}}{-j\omega} - \frac{e^{-3j\omega}-1}{j^2\omega^2}\right)$$

$$F(\omega) = \frac{3e^{3j\omega}-3}{-j\omega} - \frac{3e^{3j\omega}}{-j\omega} + \frac{e^{-3j\omega}-1}{j^2\omega^2}$$

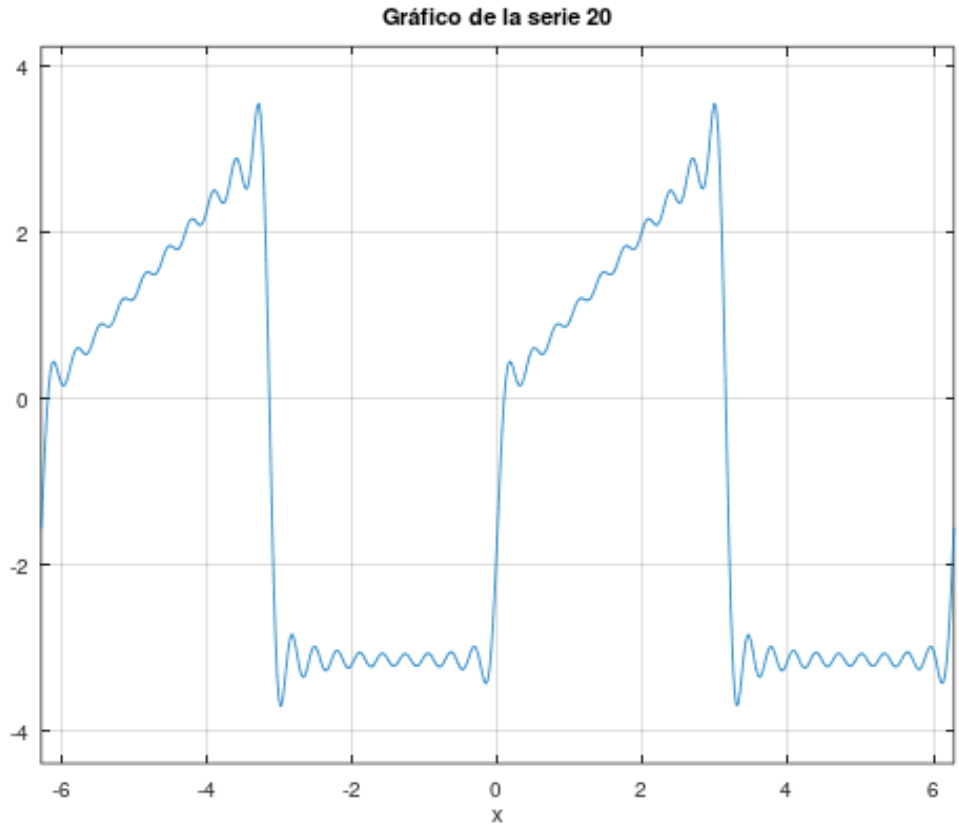
$$F(\omega) = \frac{3e^{3j\omega}-3-3e^{3j\omega}}{-j\omega} - \frac{e^{-3j\omega}-1}{\omega^2}$$

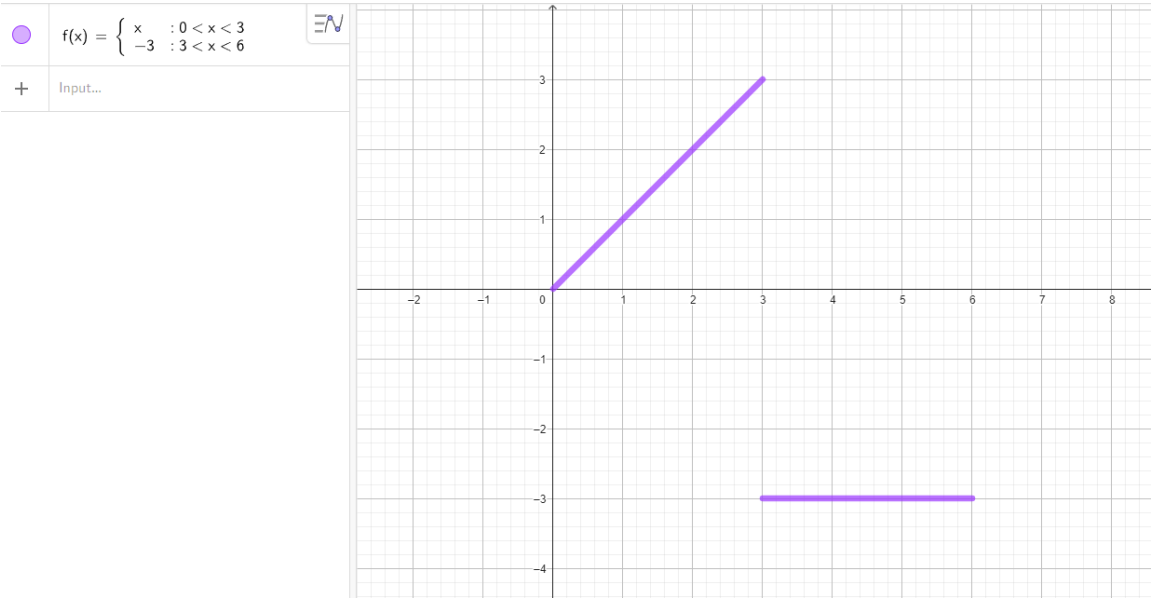
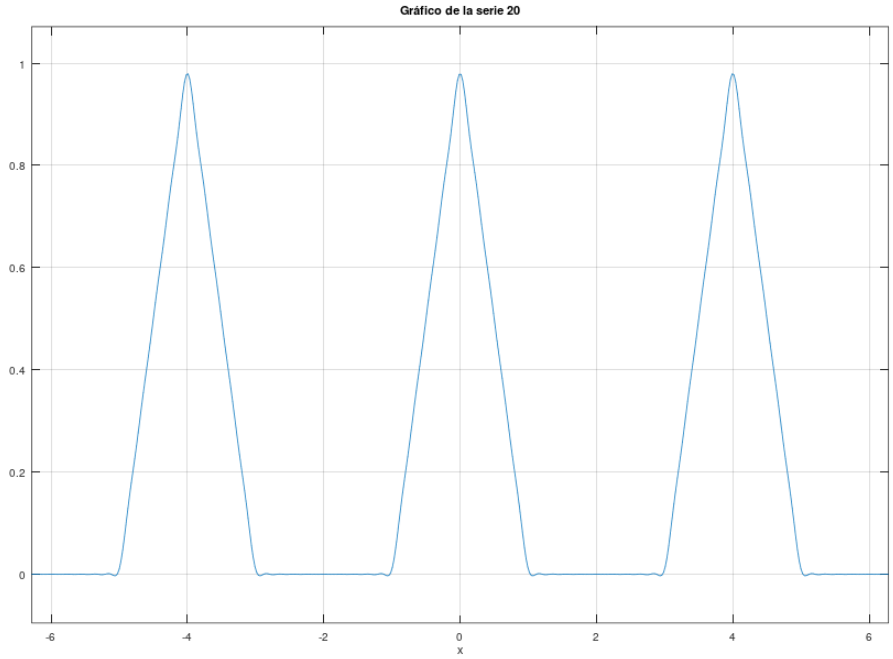
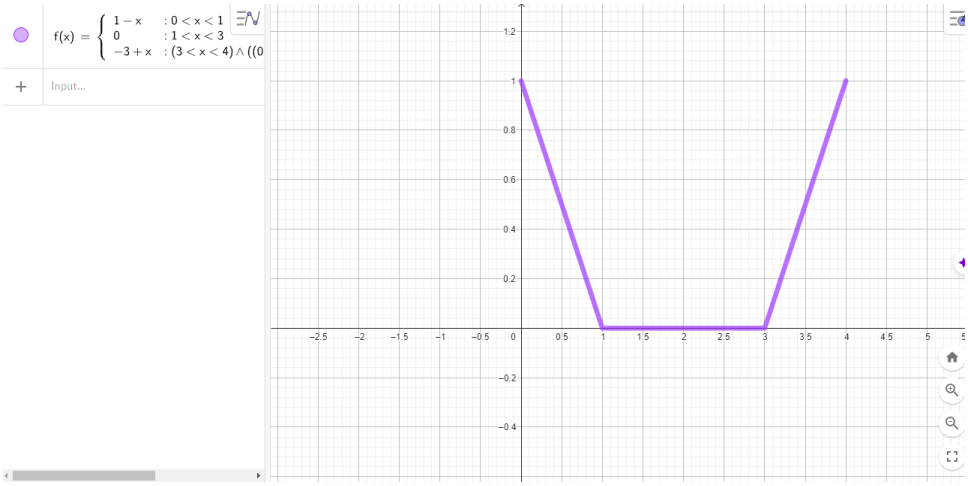
$$F(\omega) = \frac{3}{j\omega} - \frac{e^{-3j\omega}-1}{\omega^2} = \frac{-3j}{\omega} - \frac{e^{-3j\omega}-1}{\omega^2}$$

3) A) Con las funciones que se presentan, desarrolle una función que le permita graficar las sumas parciales de la serie trigonométrica respectiva. Experimente con distintos valores de n y con graficas de diferentes intervalos del eje x.

B) Utilice sus graficas para proponer una expresión de forma cerrada para una función f definida en 0 < x < L que esté representada en la serie.

1) $$f(x) = -\frac{\pi}{4} + \sum_{n=1}^{\infty}\left[\frac{(-1)^n-1}{n^2\pi}\cos nx + \frac{1-2(-1)^n}{n} sen\, nx\right]$$





2) $f(x) = \frac{1}{4} + \frac{4}{\pi^2}\sum_{n=1}^{\infty}\left[\frac{1}{n^2}\left(1 - \cos\frac{n\pi}{2}\right)\cos\frac{n\pi}{2}x\right]$